ENEL 573 – Computer Networks
Fall 2026

Chapter 1: Introduction

Instructor: Dr. Qiang (John) Ye
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Department of Electrical and Software Engineering
University of Calgary, Canada
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About Myself

- **Education**: **Ph.D. in Electrical and Computer Engineering** at the University of Waterloo, Canada, 2016
- **Work experience**
  - ✓ **Assistant Professor,** University of Calgary, Calgary, AB, Canada (Sept 2023 - Now).
  - ✓ **Assistant Professor,** Memorial University of Newfoundland, St. John's, NL, Canada (2021 - 2023).
  - ✓ **Assistant Professor,** Minnesota State University, Mankato, MN, USA (2019 - 2021).
  - ✓ **Research Associate,** University of Waterloo, Waterloo, Canada (2018 - 2019).
  - ✓ **Post-Doctoral Fellow,** University of Waterloo, Waterloo, Canada (2016 - 2018).
- **Research areas:** Network resource management for 5G/6G, **digital twin and AI assisted wireless networking**, **edge intelligence for Internet-of-Things (IoT)**, SDN/NFV, network slicing, MAC and network protocols, network performance modeling, space-air-ground integrated networking, autonomous driving, industrial/remote IoT.
- **Faculty webpage:** https://profiles.ucalgary.ca/qiang-ye
- **Personal webpage:** https://qiang-john-ye.github.io/

I am hiring!

- **Undergraduate Summer Students (USRA, PURE)**
- **MSc Students (Full Scholarships, >$25K)**

ENEL 573

Course schedule:

Lectures: MWF, 12:00pm-12:50pm, Room: BI 587

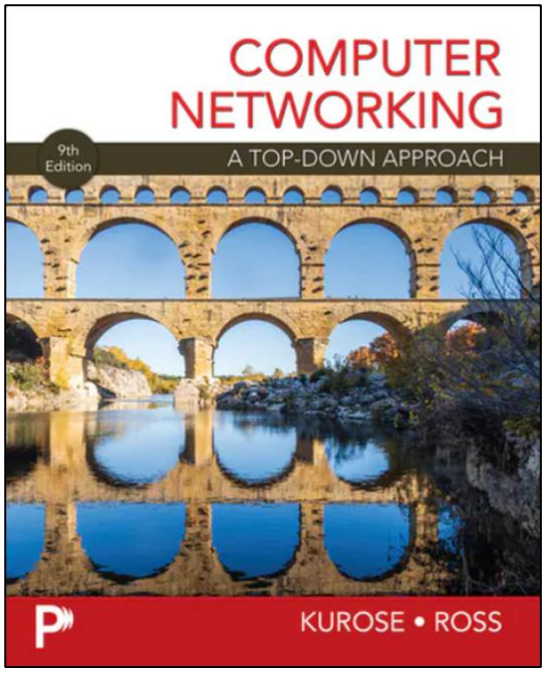
Office hours: M, 3:00pm-4:30pm, Office Room: ICT 352B

TA:

Teaching Assistant	Email	Office
Dong Jia	dong.jia@ucalgary.ca	ICT 416A
Guojun Tang	guojun.tang@ucalgary.ca	ICT 520C
Saleha Ahmed	saleha.ahmed@ucalgary.ca	ICT 416A
Muhammad Shaheer Ud Din Khan	shaheer.khan2@ucalgary.ca	ICT 359

Textbook:

James F. Kurose and Keith W. Ross, "Computer Networking: A Top-Down Approach", 9th Edition, Pearson, 2025.



Reference Books:

1) Bertsekas, D., and R. Gallager, Data Networks, Prentice Hall, 1992.
2) Sheldon M. Ross, Introduction to Probability Models, 11th Edition, Academic Press, 2014.

Evaluation

- **Assignments** (5 total): 5x5.2% = **26%**

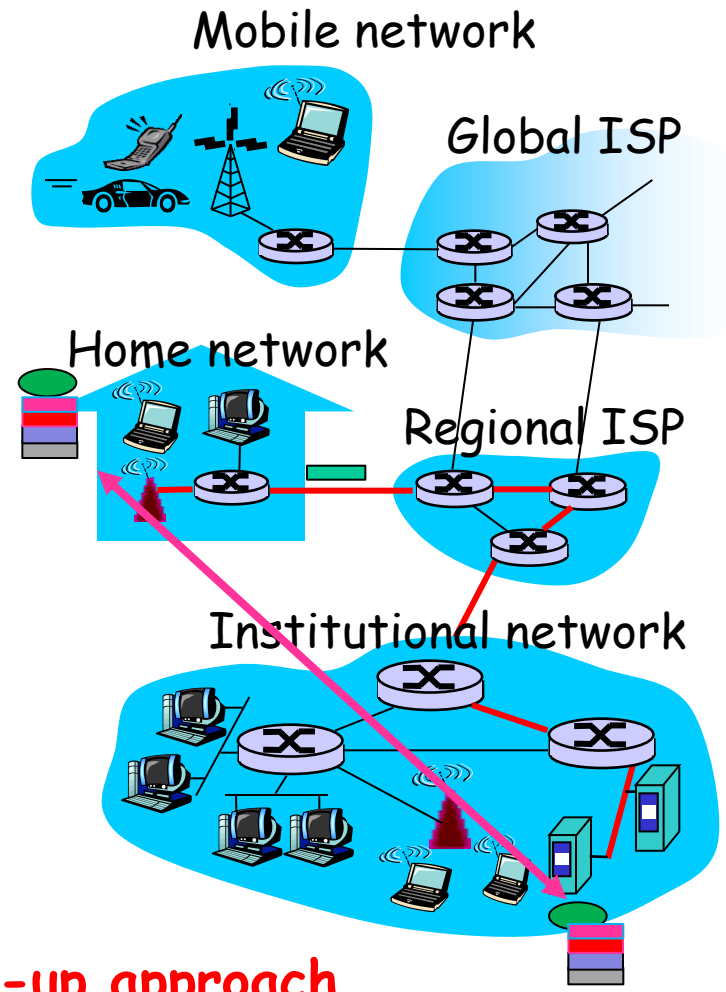
- **Projects (24%):** 3 projects that will be done by students individually
  - **PRO1**: Development of a simulator of a single transmission system (Queueing Simulation)
  - **PRO2**: Development of a simulator for retransmission protocols (Link-Layer Packet Retransmission)
  - **PRO3**: Encapsulation and network utility tools (Protocol Analysis)

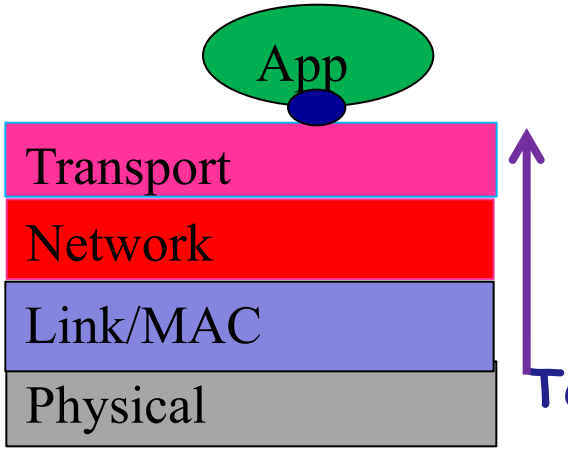
- **Midterm Exam (20%)**
  - **Date & Time: Oct. 26 (Monday)** during the class time from **12:00pm to 12:50pm**
  - **Location: ENA 101**

- **Final Exam (30%)**
  - Duration: 3hrs
  - Time & Location: TBD

Objectives of The Course

1. To understand the fundamental concepts in computer networks
2. To understand the principles and practice of designing, analyzing, and operating networks.





Teaching: Bottom-up approach

Outline of the course

- Chap1 Introduction: Definition of networks, network edge, network core, how it works, performance
- Chap2 Network Analysis and Queuing Theory: Randomness in communication networks, A review of probability theory, Queuing theory
- Chap3 Data link layer: Error detection and correction, Retransmission, Medium access control (MAC) protocols: Aloha, CSMA/CD and CSMA/CA.
- Chap4 Network layer: IP addressing, fragmentation, Routing
- Chap5 Transport layer: TCP and UDP, Flow control and Congestion control
- Chap6 Enabling technologies for 5G networks and data and network security

A note about the slides

Some of the slides were originally prepared by Drs. Kurose and Ross.

I have added more slides and edited some.

Chapter 1: Introduction

1.1 What is the Internet?

1.2 Network edge

- end systems, access networks, links

1.3 Network core

- circuit switching, packet switching, network structure

1.4 The way networks work

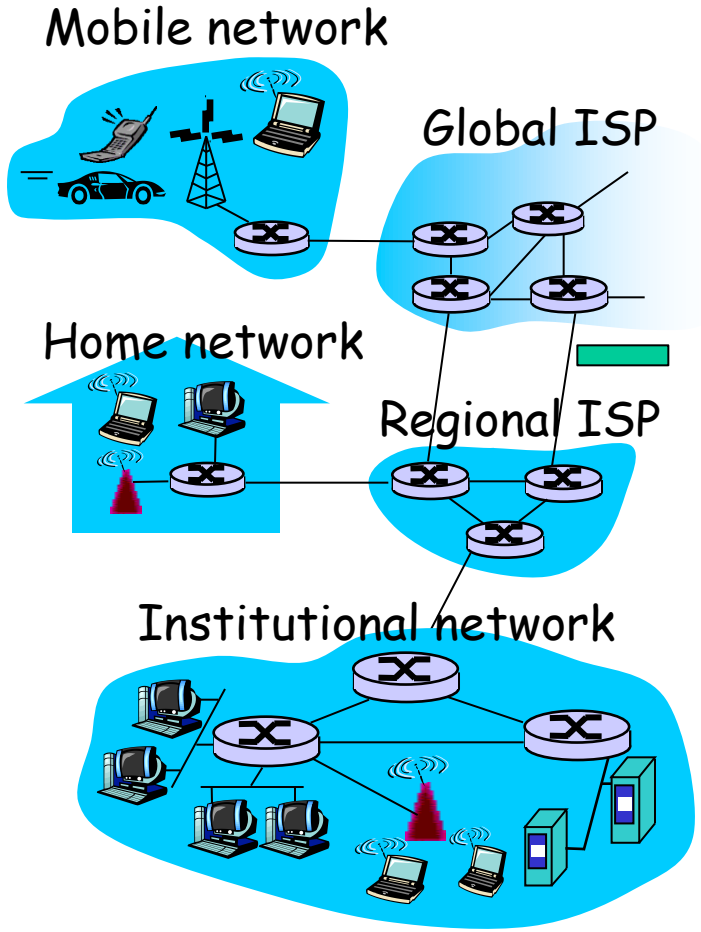
1.5 Performance metrics:

Delay, loss and throughput in packet-switched networks

What's the Internet: "nuts and bolts" view



- Millions/Billions of connected computing devices: hosts = end systems
  running network apps
- Communication links
  - fiber, copper, radio, satellite
  - Bandwidth (transmission rate)
- BSs/APs/Routers: forward packets (chunks of data)



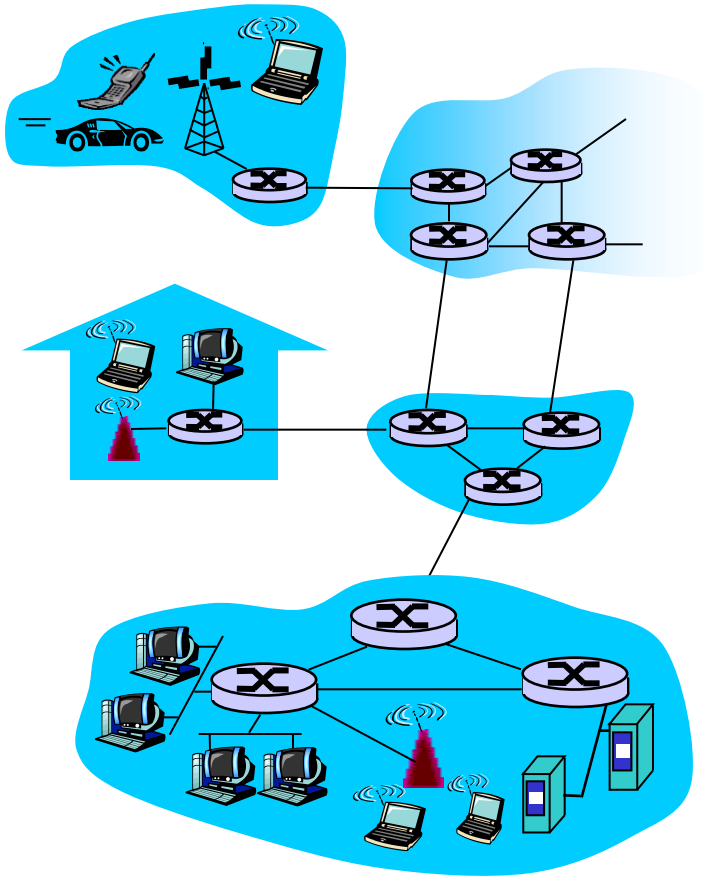
What's the Internet: "nuts and bolts" view

- Internet: "network of networks"
  - loosely hierarchical
- Internet permits the reliable exchange of information with low cost
- Protocols control sending, receiving of messages
  - e.g., TCP, IP, HTTP, Ethernet
- Internet standards
  - RFC: Request For Comments
  - IETF: Internet Engineering Task Force

What's the Internet: a service view

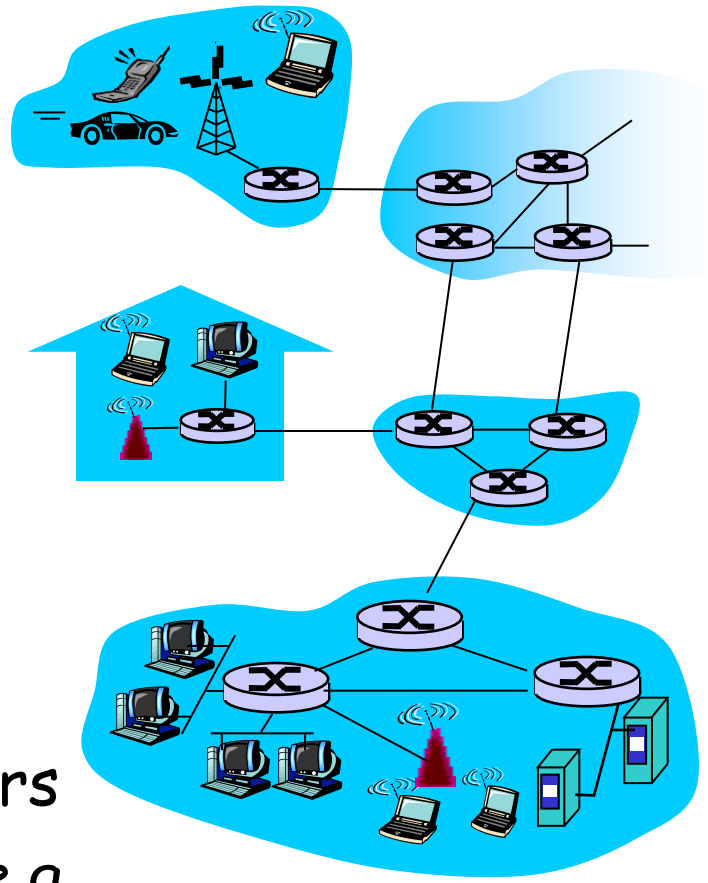
- Communication infrastructure enables distributed applications:
  - Web, VoIP, email, online-games,
  - e-commerce, file sharing

- Communication services provided to apps:
  - reliable data delivery from source to destination
  - "best effort" (unreliable) data delivery



A closer look at network architecture:

- Network edge:
  - Applications and hosts
  - Access networks
  - Physical media: wired, wireless communication links



- Network core:
  - Interconnected routers/servers
  - Advanced network functions (e.g., access/mobility management)

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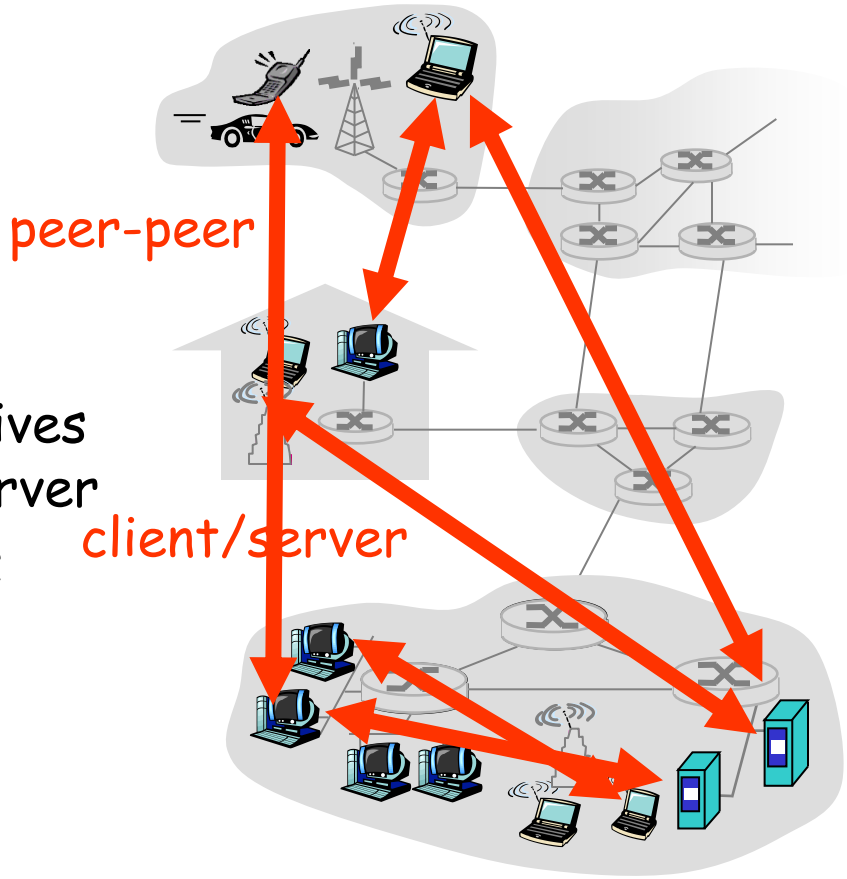
1.4 The way networks work

1.5 Performance metrics:

Delay, loss and throughput in packet-switched networks

The network edge:

- End systems (hosts):
  - run application programs
  - e.g., Web, email
  - at "edge of network"
- Client/server model
  - client host requests, receives service from always-on server
  - e.g., Web browser/server; email client/server
- Peer-peer model:
  - minimal (or no) use of dedicated servers
  - e.g., Skype, Twitter, Tiktok, Facebook



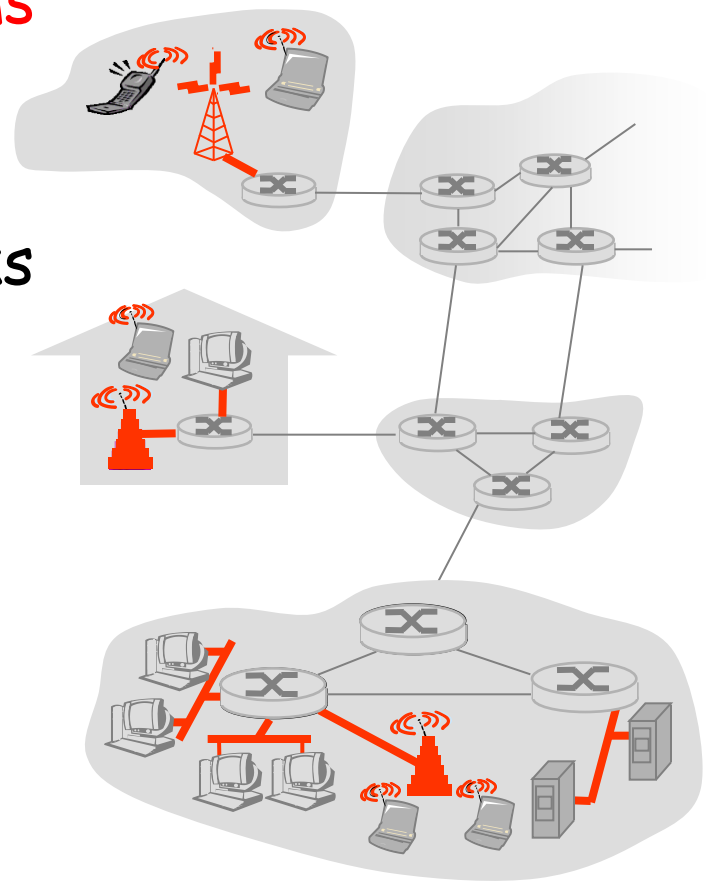
Access networks and physical media

Q: How to connect end systems to edge router?

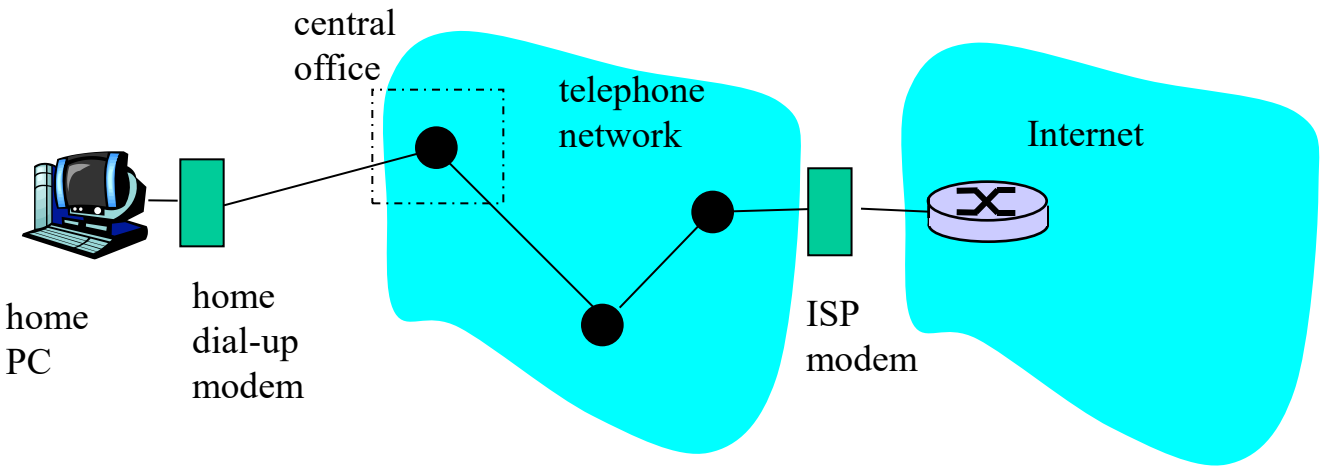
- residential access nets
- institutional access networks (school, company)
- mobile access networks

Keep in mind:

- bandwidth (bits per second) of access network?
- shared or dedicated?

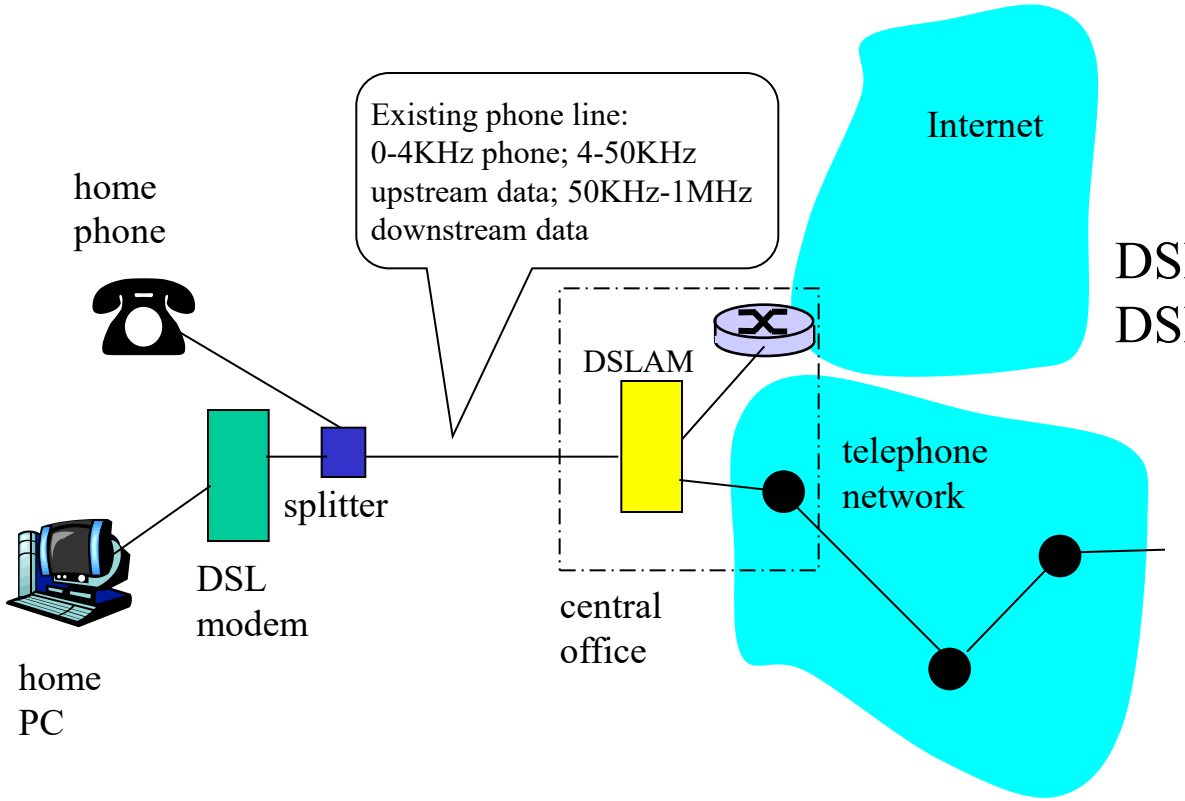


Dial-up Modem



- uses existing telephony infrastructure
  - home directly-connected to central office
- up to 56Kbps direct access to router (often less)
- can't surf, phone at same time: not "always on"

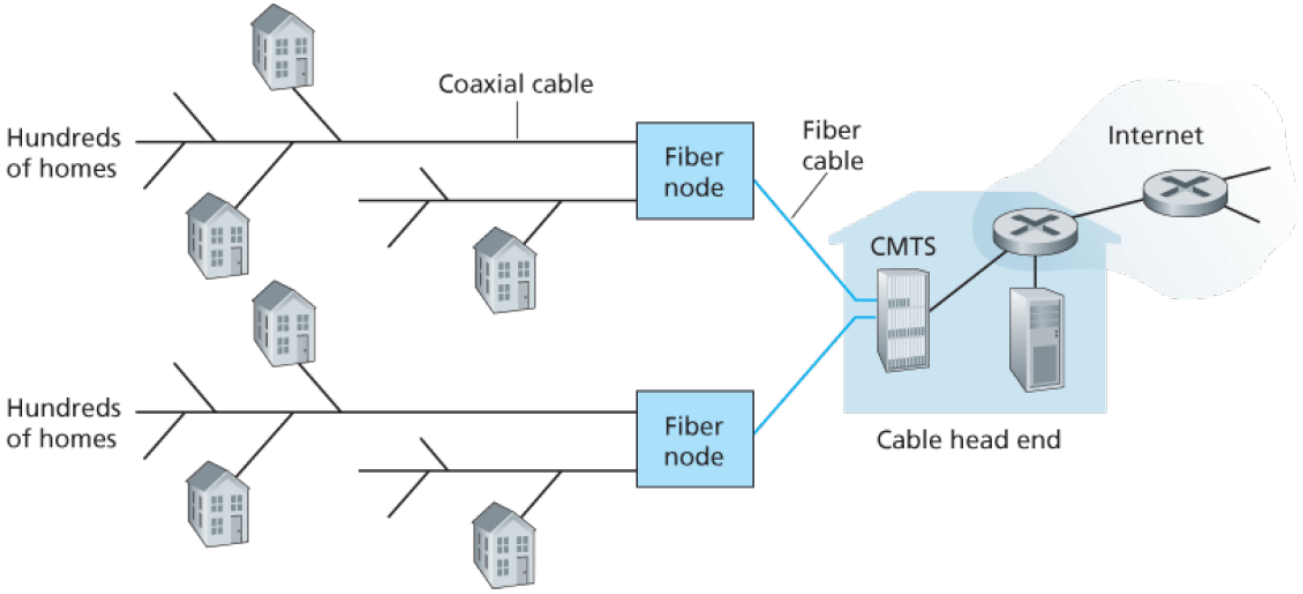
Digital Subscriber Line (DSL)



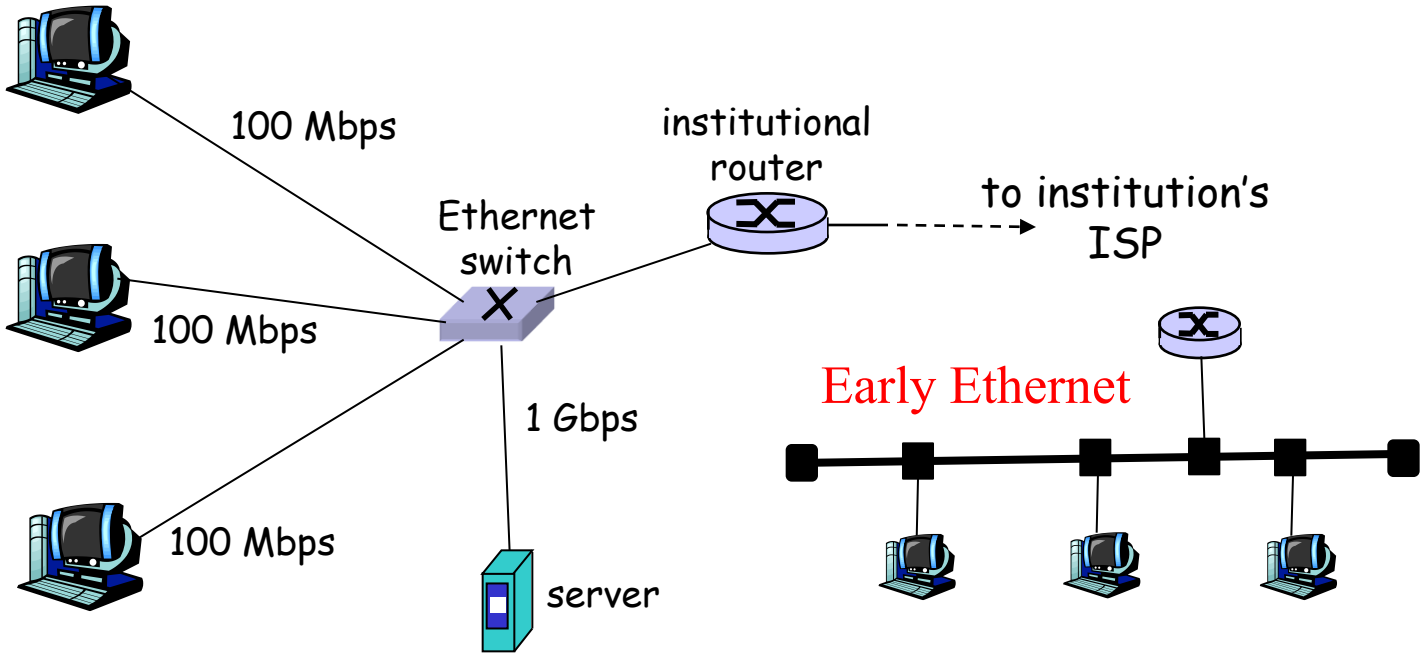
- uses existing telephone infrastructure
- up to 4k-50kbps upstream (today typically < 256 kbps)
- up to 50k-1Mbps downstream (today typically < 1 Mbps)
- dedicated physical line to telephone central office

Residential access: cable modems

- uses cable TV infrastructure
- HFC: hybrid fiber coax
  - asymmetric: up to 30Mbps downstream, 2 Mbps upstream. Now, it is gigabit-class broadband
- Network of cable, fiber attaches homes to ISP router
  - homes share access to router
  - unlike DSL, which has dedicated access



Ethernet Internet access

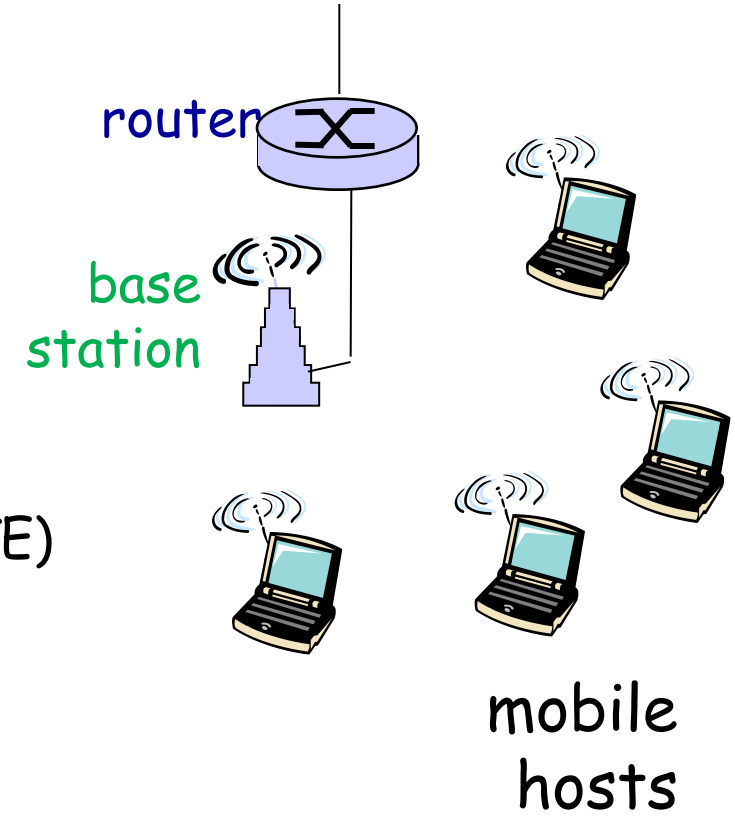


- typically used in companies, universities, etc
- 10 Mbps, 100Mbps, 1Gbps, 10Gbps, 100G Ethernet
- Question: How do nodes efficiently share the medium?

CSMA/CD (Carrier Sense Multiple Access with Collision Detection)

Wireless access networks

- Shared wireless access network connects end system to router
  - via base station aka "access point"
- Wireless LANs:
  - 802.11b/g/n/ac/ax (WiFi)
- Wider-area wireless access
  - provided by telco operator
  - ~100Mbps over cellular system (LTE)
  - NB-IoT
  - LoRaWAN (LoRa)



Physical Media

- bit: propagates between transmitter/receiver pairs
- physical link: what lies between transmitter & receiver
- guided media:
  - signals propagate in solid media: copper, fiber, coax
- unguided media:
  - signals propagate freely, e.g., radio

Twisted Pair (TP)

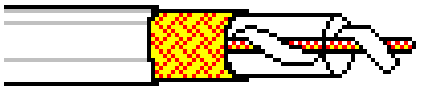
- two insulated copper wires
  - Category 3: traditional phone wires, 10 Mbps Ethernet
  - Category 5: 100Mbps Ethernet
  - Category 5e: 1 Gbps Ethernet
  - Category 6: 10 Gbps Ethernet



Physical Media: coax, fiber

Coaxial cable:

- two concentric copper conductors
- bidirectional
- baseband:
  - single channel on cable
  - legacy Ethernet
- broadband:
  - multiple channels on cable
  - HFC



Fiber optic cable:

- glass fiber carrying light pulses, each pulse a bit
- high-speed operation:
  - high-speed point-to-point transmission (e.g., 10's-100's Gbps)
- low error rate: repeaters spaced far apart ; immune to electromagnetic noise



Physical media:

Radio:

- signal carried in electromagnetic spectrum
- no physical "wire"
- bidirectional
- propagation environment effects:
  - reflection
  - obstruction by objects
  - interference

Radio link types:

- terrestrial microwave
  - e.g., up to Gbits capacity
- LAN (e.g., WiFi)
  - 11Mbps, 54 Mbps, ~1Gbits
- wide-area (e.g., cellular)
  - 4G cellular: ~ 10 Mbps, ~1Gbits
- satellite
  - 10-100Mbps channel (or multiple smaller channels)
  - 100 msec end-end delay
  - geosynchronous versus low altitude

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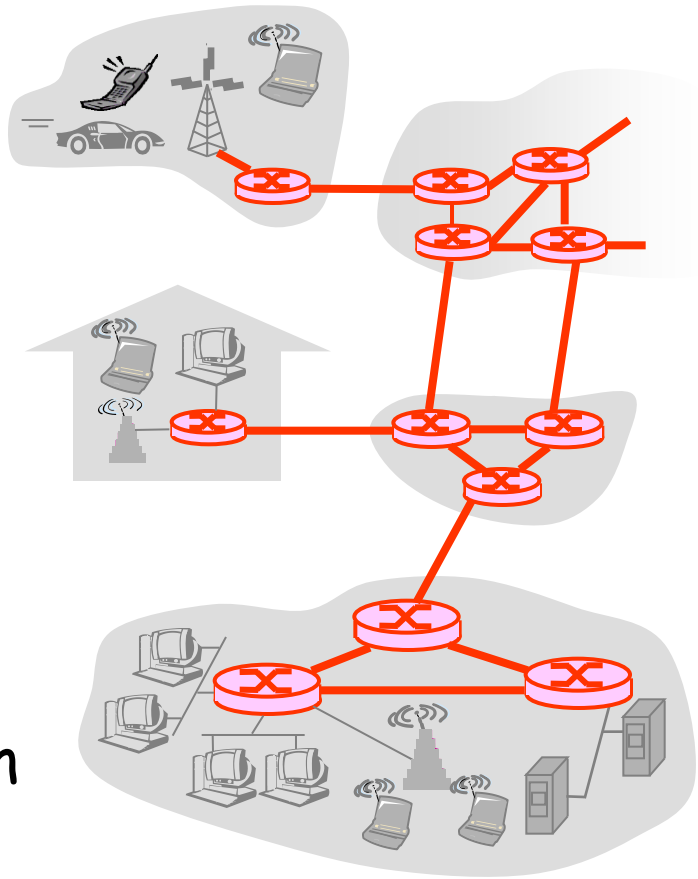
1.5 Performance metrics:

Delay, loss and throughput in packet-switched networks

The Network Core

- Mesh of interconnected routers
- the fundamental question:

  how is data transferred through net?
  - Circuit-switching: dedicated circuit per call (telephone net)
  - Packet-switching: packets can be transmitted following different paths in the network, and the resources can be shared by other connections. (Internet)



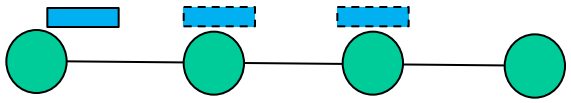
Network Core: Packet Switching

Each end-end data stream divided into packets

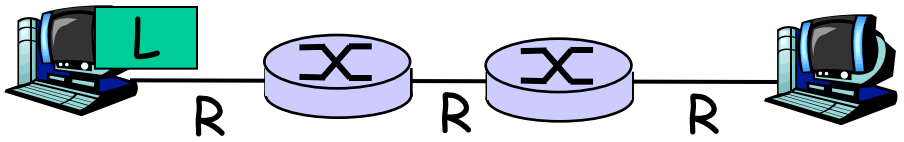
- user A, B packets share network resources
- each packet uses full link bandwidth
- resources used as needed

Resource contention:

- aggregate resource demand can exceed amount available
- congestion: packets queue, wait for link use
- store and forward: packets move one hop at a time
  - node receives complete packet before forwarding



Packet-switching: store-and-forward



- takes L/R seconds to transmit (push out) packet of L bits onto link at R bps
- store and forward: entire packet must arrive at router before it can be transmitted on next link
- delay = 3L/R (assuming zero propagation delay)

more on delay shortly …

Example:

- L = 7.5 Mbits

  (Note: packets are not that long! ~1.5KBytes is very common)
- R = 1.5 Mbps
- transmission delay = 5 sec

Network Core: Packet Switching

- A message is segmented into blocks of data called packets. A packet is a group of bits, typically from a few hundreds to thousands.



- A packet consists of a header, user information, and a trailer. The header usually contains the addresses of the destination and of the source of the packet; it may also include a sequence number that the destination users to verify that all the packets were received or to reorder them. The trailer contains error control bits that the nodes use to verify that they received the packet correctly.

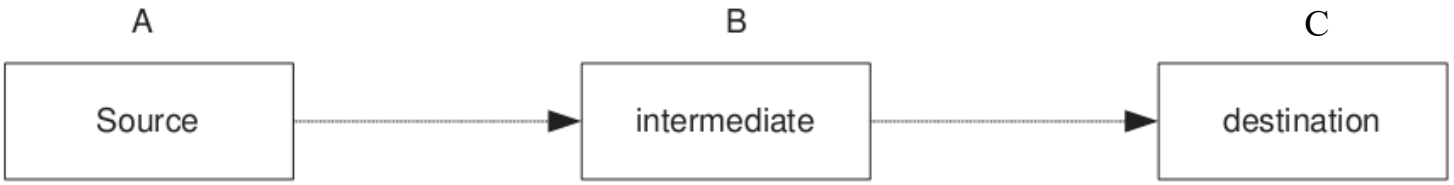
Packet Switching

- Packets are transmitted without prior reservation of link capacity. When a packet is received at a switch, it is inspected to determine the appropriate output link. If the output link is available, it is transmitted. Otherwise, it is stored and then forwarded to the next switch on its way to the destination (store & forward network)

- The transmission of messages as small packets is called store-and-forward packet switching. The store-and-forward packet switching can reduce the message delivery time. The reduction in the delivery time is called the pipelining gain .

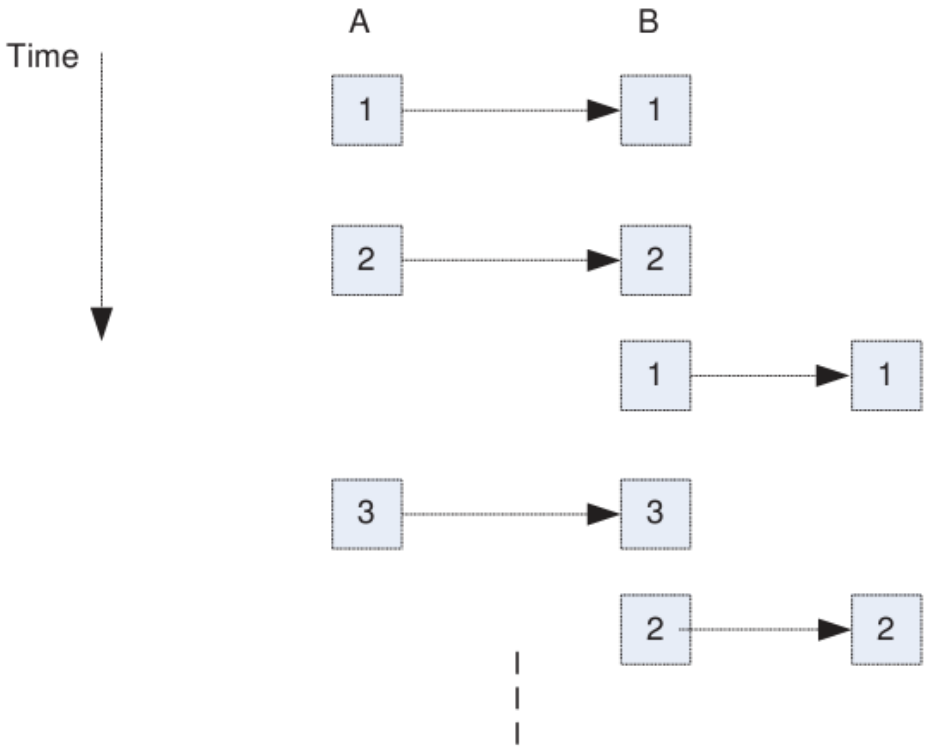
Packet Switching

- Example: Transmission of a message on a direct link from A to B or B to C takes 1 minute. The total transmission time takes 2 minutes.



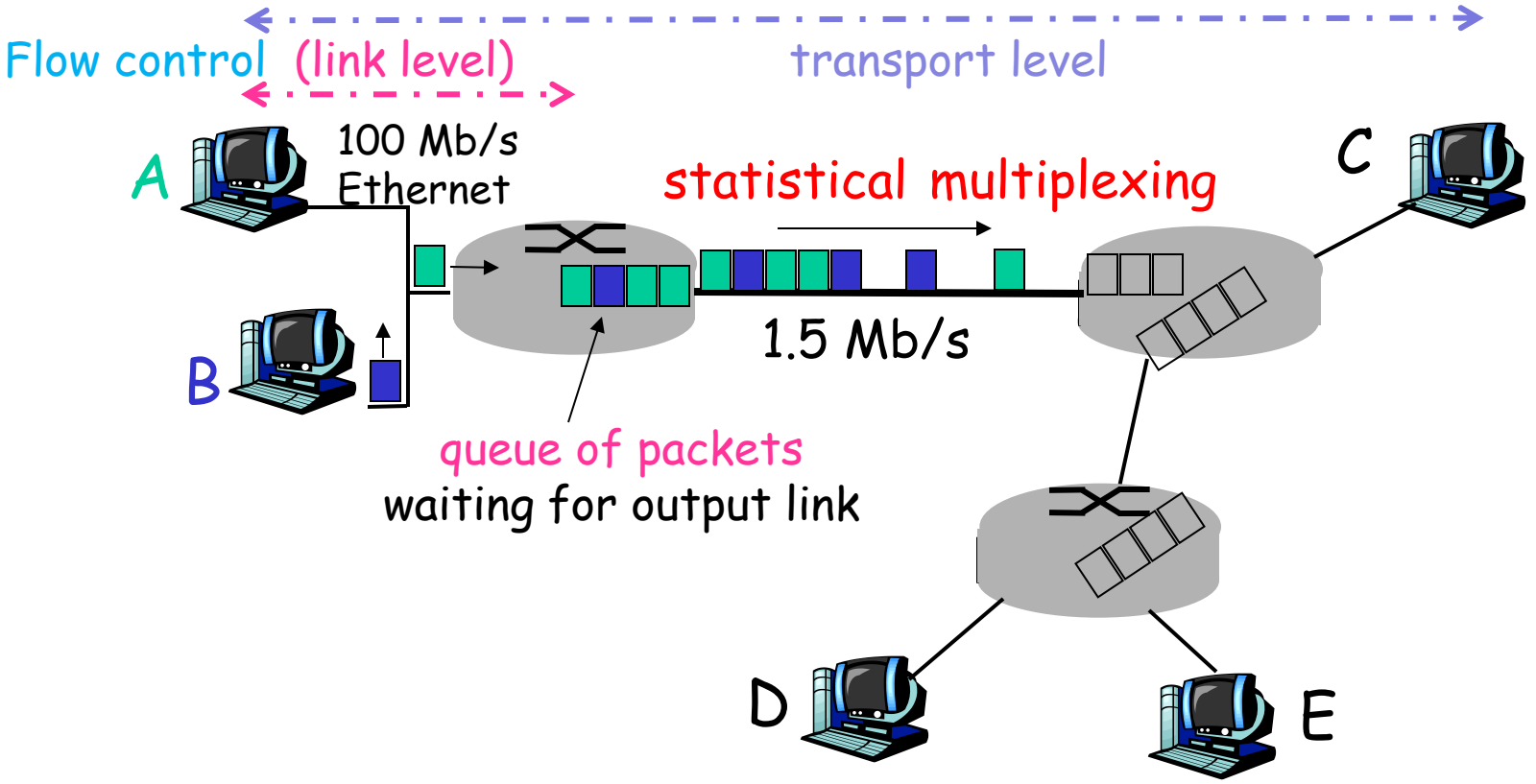
If the message is decomposed into 60 packets that takes 1 second each to be transmitted on a direct link, then during the first second packet 1 is sent from A to B, during the next second packet 2 is sent from A to B and packet 1 is sent from B to C, and so on. After 1 minute and 1 second, the complete message is received by C .

Packet Switching



If there are N intermediate nodes, the transmission of the undivided message takes N+1 minutes while that of the 60 packets takes only 1 minute and N seconds.

Packet Switching: Statistical Multiplexing

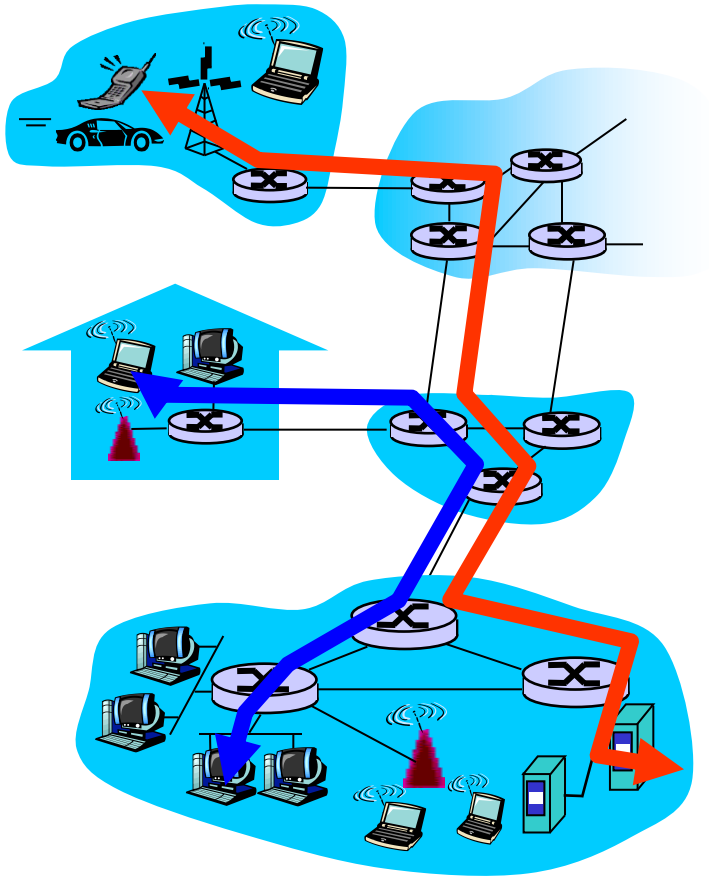


- sequence of A & B packets has no fixed timing pattern
  - bandwidth shared on demand: statistical multiplexing.
- TDM: each host gets same slot in revolving TDM frame.

Network Core: Circuit Switching

End-end resources reserved for "call"

- link bandwidth, switch capacity
- dedicated resources: no sharing
- circuit-like (guaranteed) performance
- call setup required



Network Core: Circuit Switching

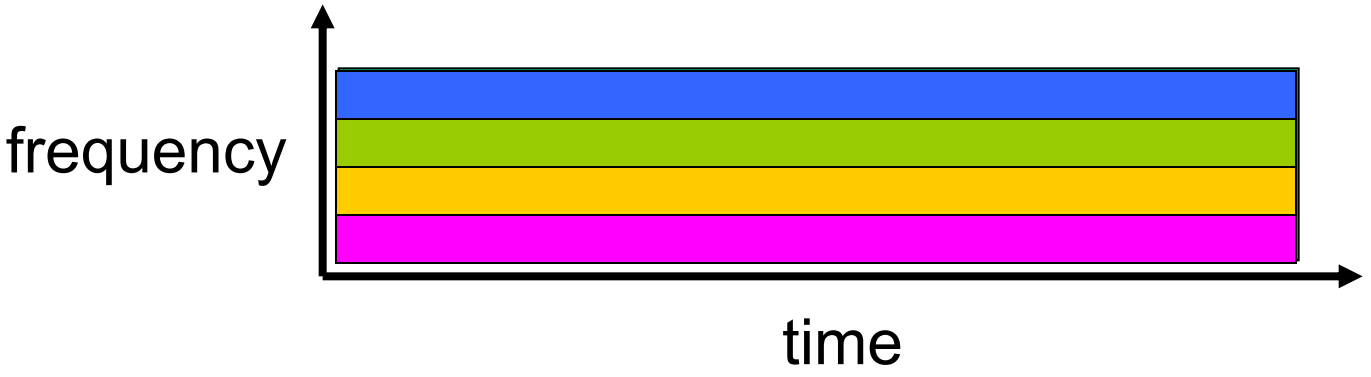
Network resources (e.g., bandwidth) divided into "pieces"

- pieces allocated to calls
- resource piece idle if not used by owning call (no sharing)
  - dividing link bandwidth into "pieces"
    - frequency division
    - time division

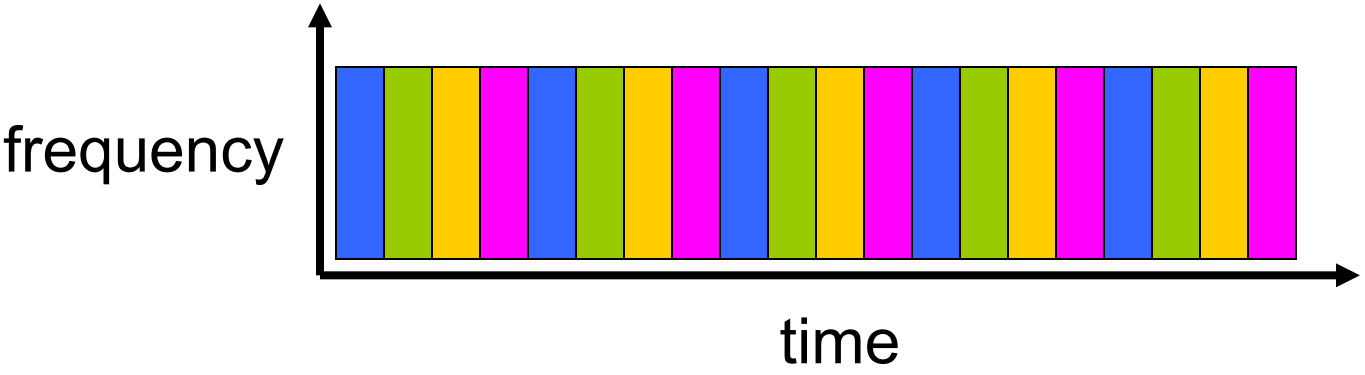
Circuit Switching: FDM and TDM

Example: 4 users 

FDM:



TDM:



Packet switching versus circuit switching

Packet switching allows more users to use network!

Example:

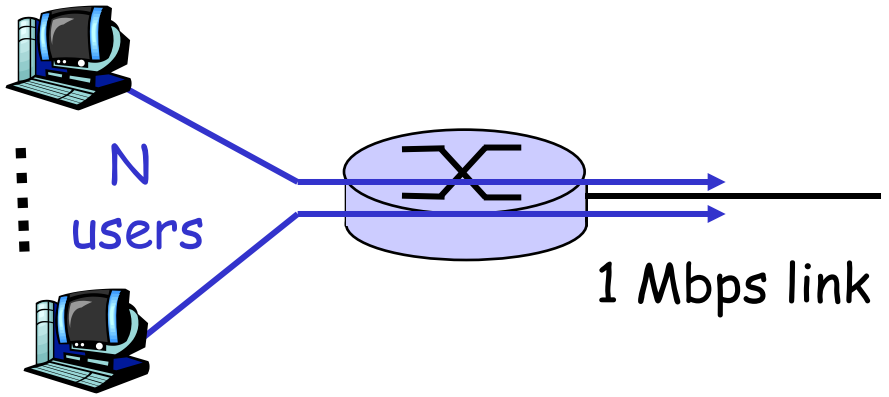
- 1 Mb/s link
- each user:
  - 100 kb/s when "active"
  - active 10% of time

❖ circuit-switching:

- 10 users

❖ packet switching:

- with 35 users, probability > 10% active at same time is 0.0004



A comparison of circuit-switched and packet-switched networks

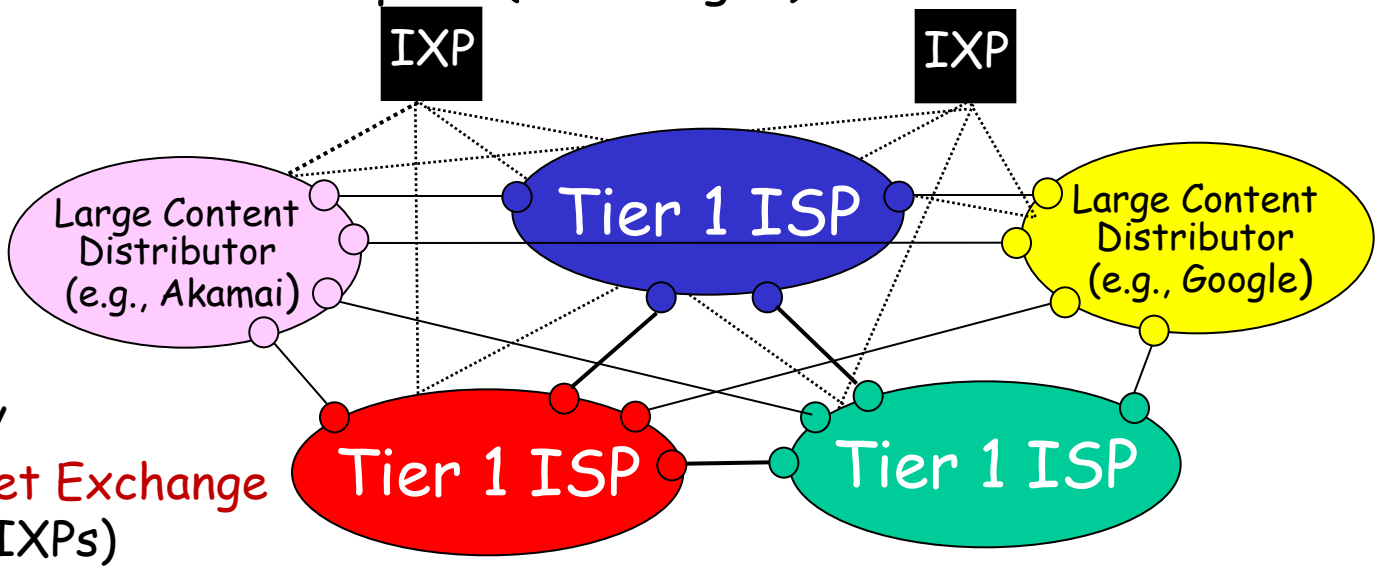
Item	Circuit switching	Packet switching
Dedicated path	Yes	No
Bandwidth available	Fixed	Dynamic
Potentially wasted	Yes	No
Store-and-forward trans.	No	Yes
Call setup	Yes	Not needed
When can congestion occur	At setup time	On every packet
Charging	Per minute	Per packet
Each packet follows the same route	Yes	No

Internet structure: network of networks

- roughly hierarchical
- at center: small # of well-connected large networks
  - "tier-1" commercial ISPs (e.g., Verizon, Sprint, AT&T, Qwest, Level3), national & international coverage
  - large content distributors (Google, Akamai (Netflix: Open Connect), Microsoft)
  - treat each other as equals (no charges)

Tier-1 ISPs & Content Distributors, interconnect (peer) privately

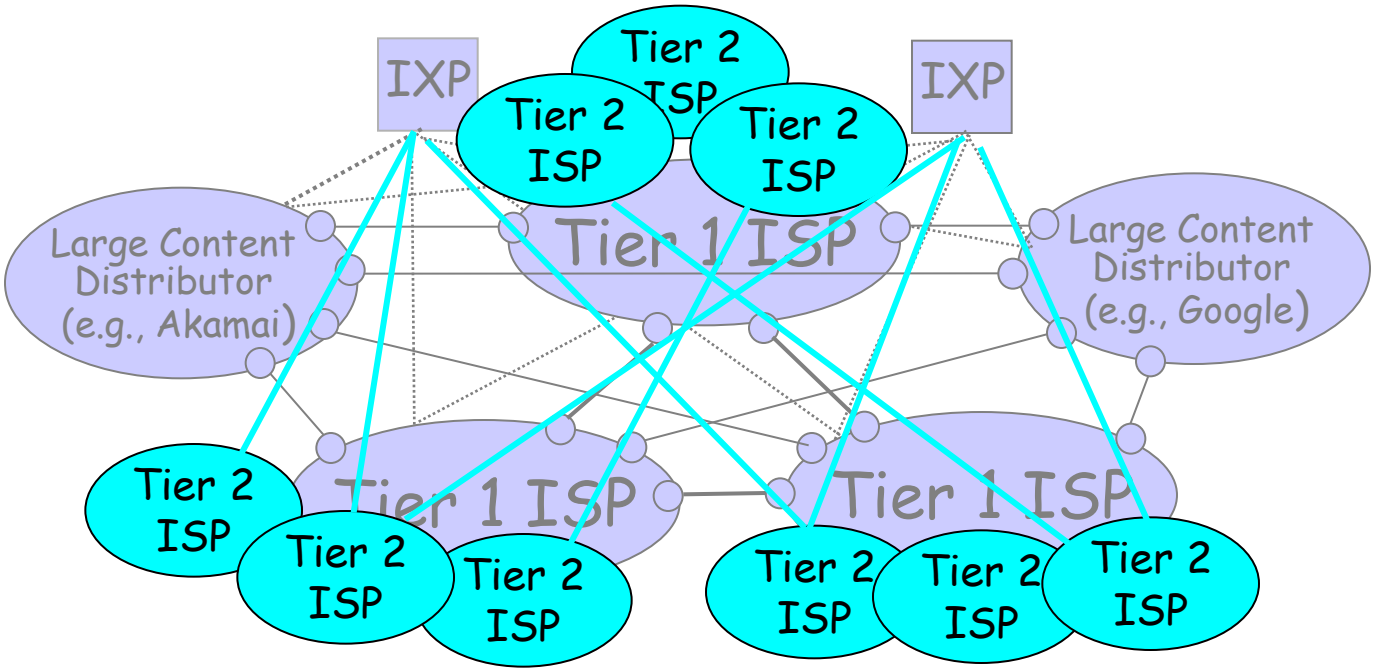
… or at Internet Exchange Points (IXPs)



Internet structure: network of networks

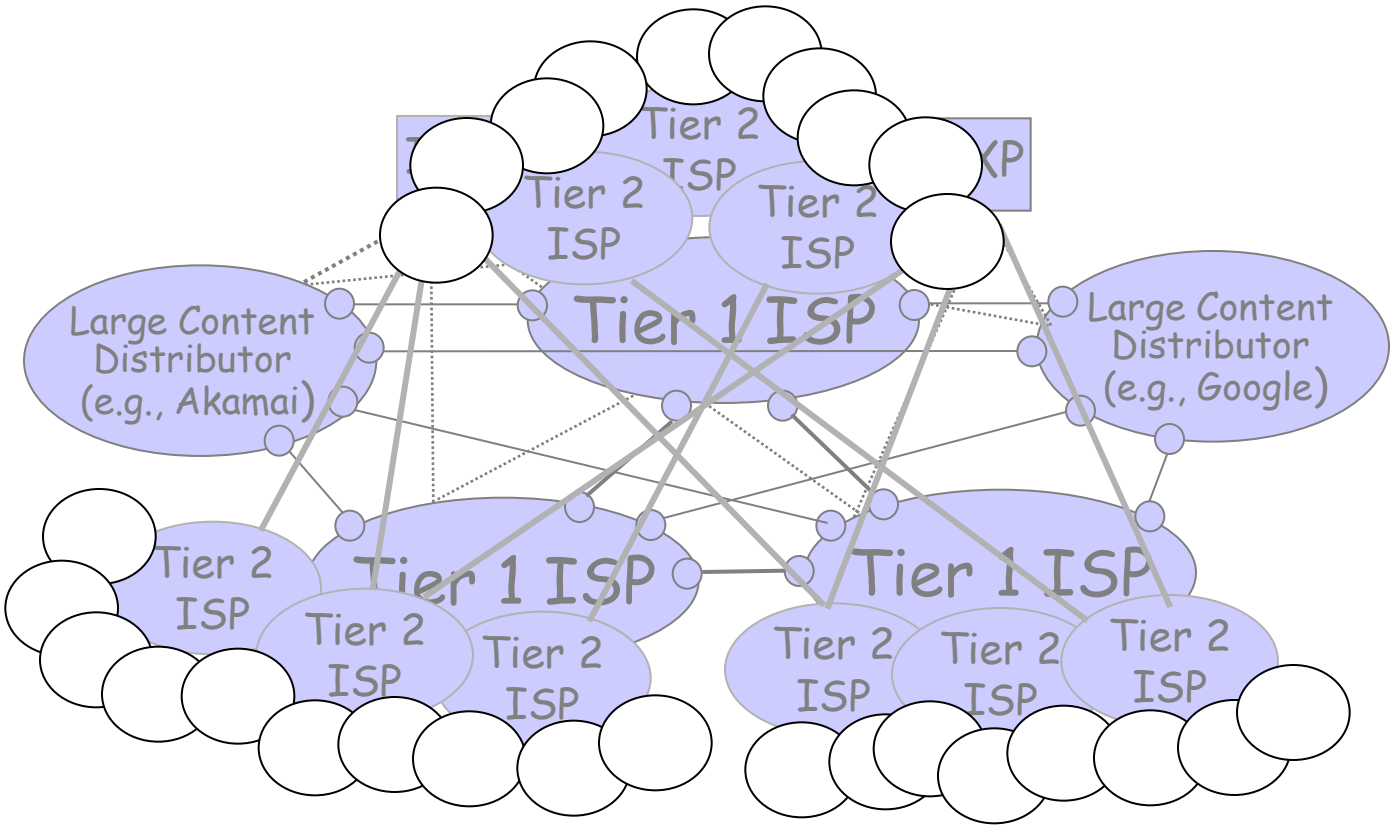
"tier-2" ISPs: smaller (often regional) ISPs

- connect to one or more tier-1 (provider) ISPs
  - each tier-1 has many tier-2 customer nets
  - tier 2 pays tier 1 provider
- tier-2 nets sometimes peer directly with each other (bypassing tier 1), or at IXP



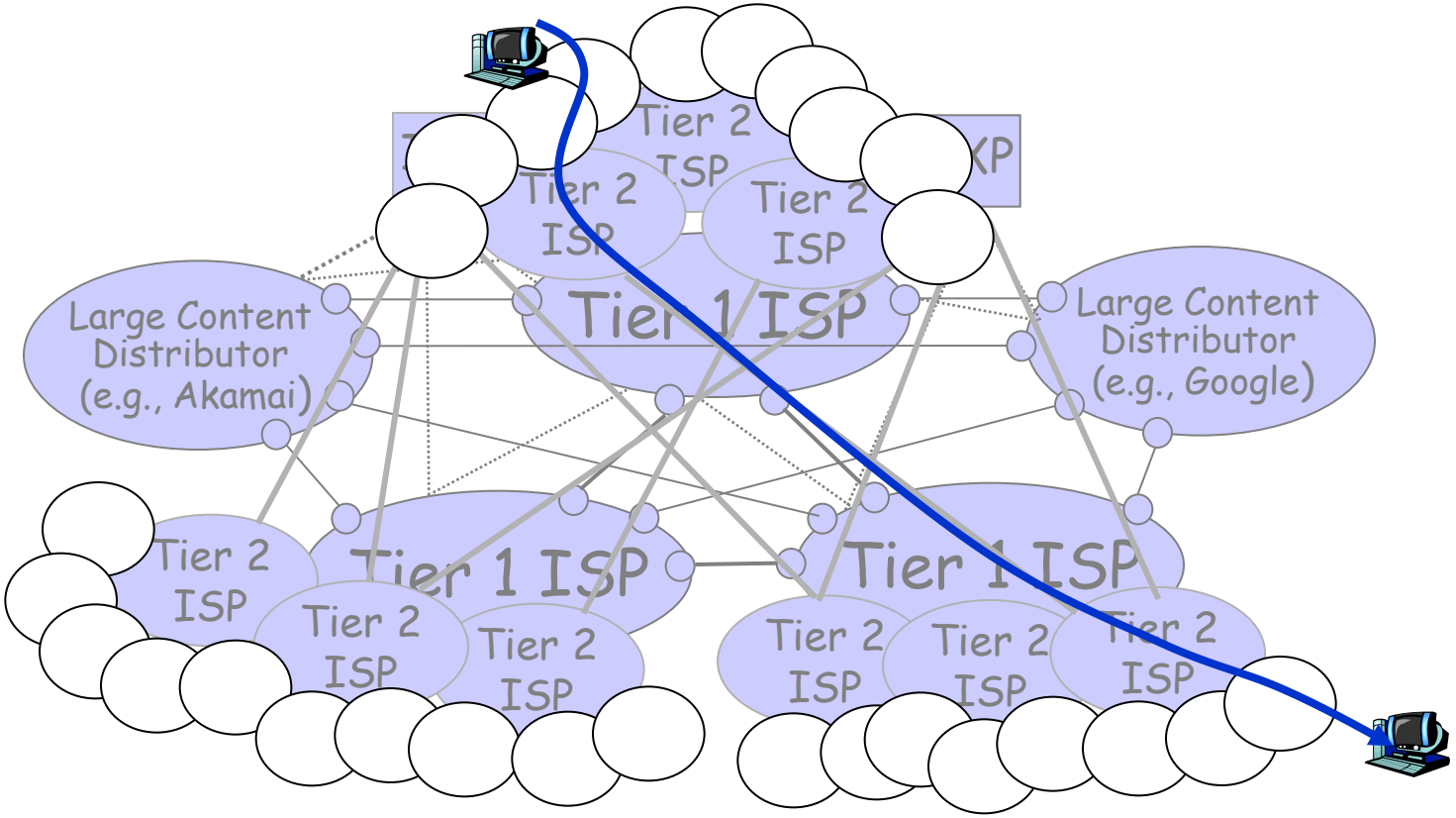
Internet structure: network of networks

- "Tier-3" ISPs, local ISPs
- customer of tier 1 or tier 2 network
  - last hop ("access") network (closest to end systems)



Internet structure: network of networks

- a packet passes through many networks from source host to destination host



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The way networks work

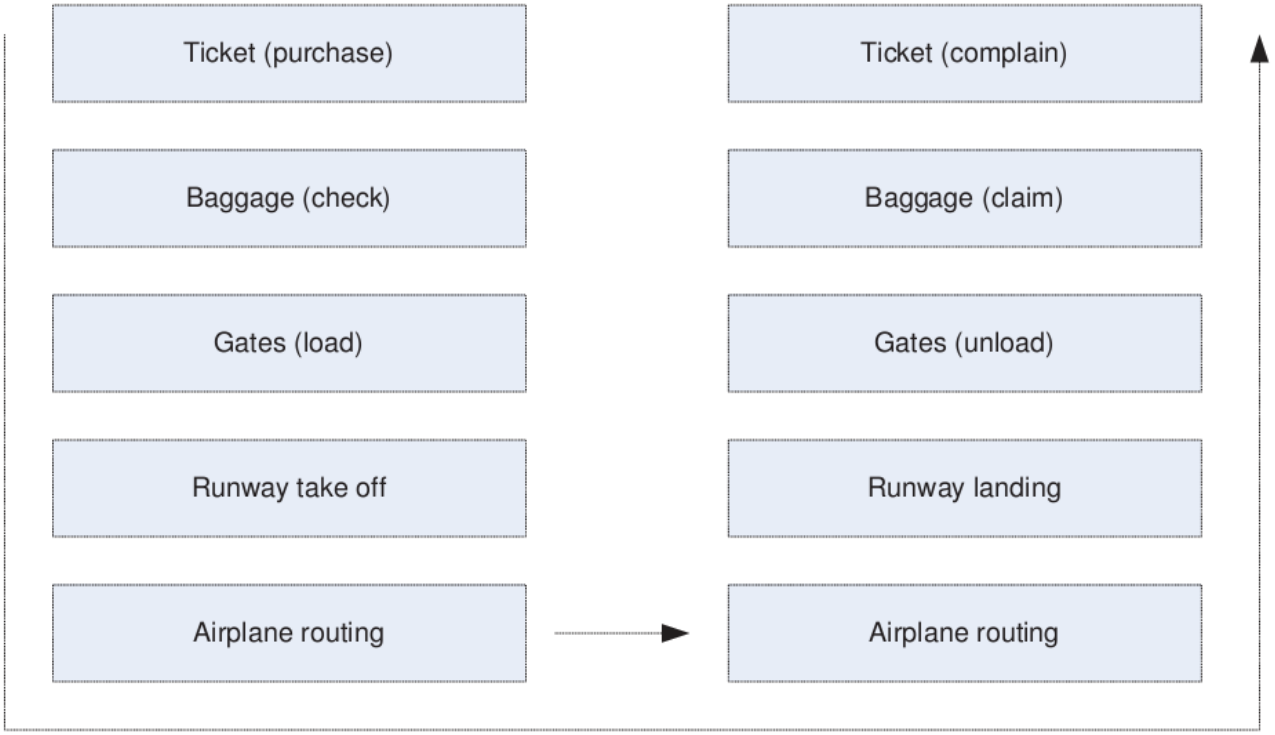
- Network functions are organized in a layered architecture. The services of one layer are implemented on top of the services provided by the layer immediately below.

- The different layers can be designed more or less independently, which greatly simplifies network design.

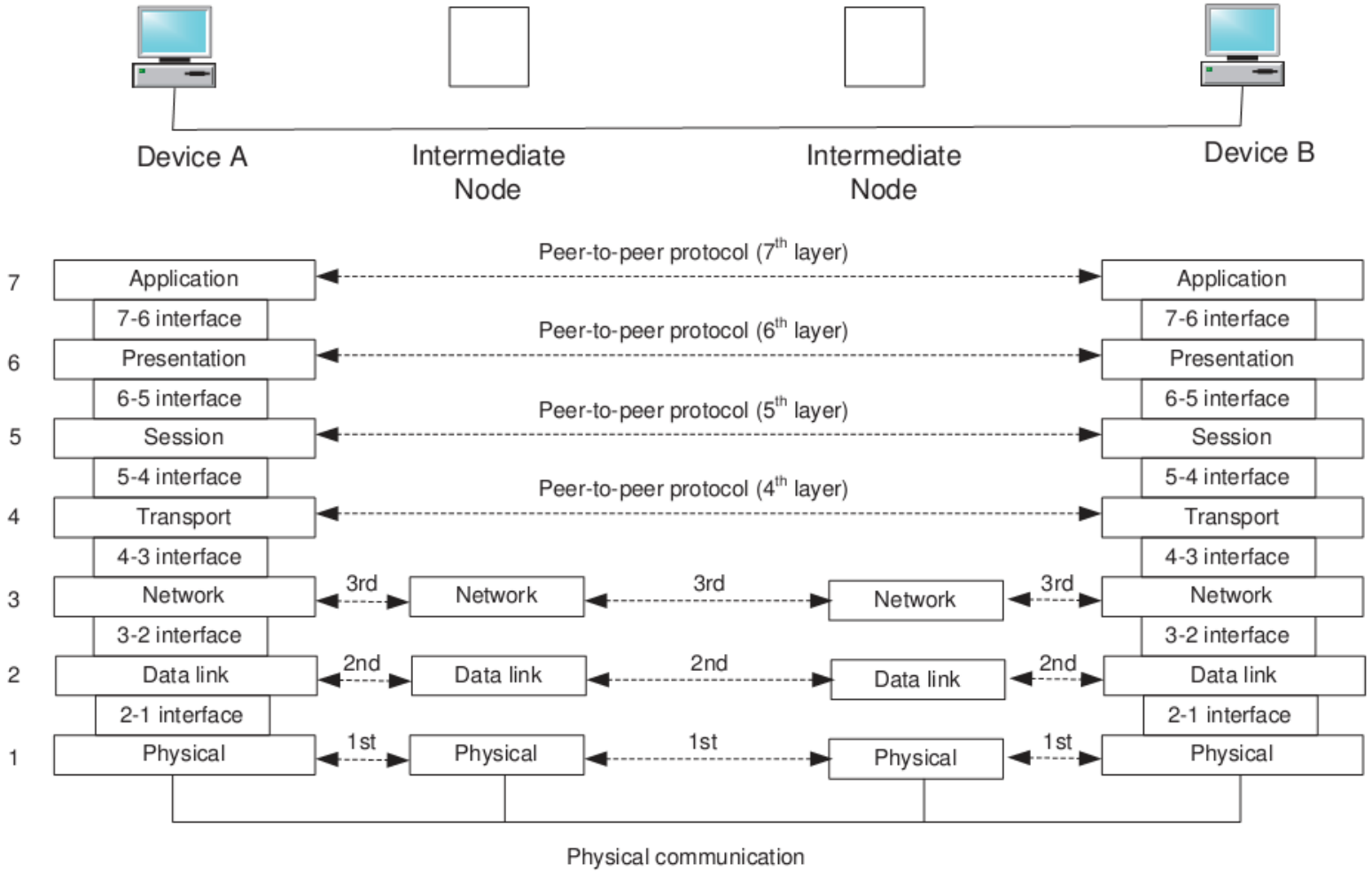
- Another advantage is the compatibility derived from the independence of the layers. For instance, when different networks are interconnected, a computer on one network can access computers on all the networks, independently of the specific implementations of the different networks.

The Way Networks Work

The operation is organized in a layered architecture. Each layer implements a service via its own internal-layer actions relying on services provided by layer below.



- The International Organization for Standardization (ISO) proposed the open systems interconnection reference model (OSI model), which is a layered architecture with seven layers.



Definitions

- Protocol: a set of rules that governs communications.
  (A protocol defines what is communicated, how and when it is communicated);
- Layering: it achieves a functional level of modularity;
- Protocol stack: a set of layers and protocols;
- Peer-to-peer protocols: protocols which make the layer N of the source and destination conceptual understand
- Interface: it defines what information and services a layer must provide for the layer above it

Note:

- Peer-to-peer protocol is achieved using the service provided by their lower layer entities
- Communication functions are partitioned into a vertical set of layers
- Ideally, upper layer entities are independent of the details of the lower layer entities, this is easy to modify protocol at one layer, independent of other layers.
- Allow heterogeneous users to communicate (open system)

Functions of layers

- Application layer implements commonly used communication services e.g., file transfer, directory services, virtual terminal.
- Presentation layer takes care of data compression, security, and format conversions so that nodes using different representations of information can communicate efficiently and securely.
- Session layer uses the transport layer services to set up and supervise connections between end systems
- Transport layer supervises end-to-end transmission of packets. This layer may arrange for retransmission of erroneous packets.
- Network layer guides the packets from their source to their destination, along a path that may comprise a number of links
- Data link layer provides reliable transmission between nodes that are attached to the same physical link.
- Physical layer transmits raw bit stream over channel

Summary of the functions of layers

The seven layers can be thought of as belonging to three subgroups.

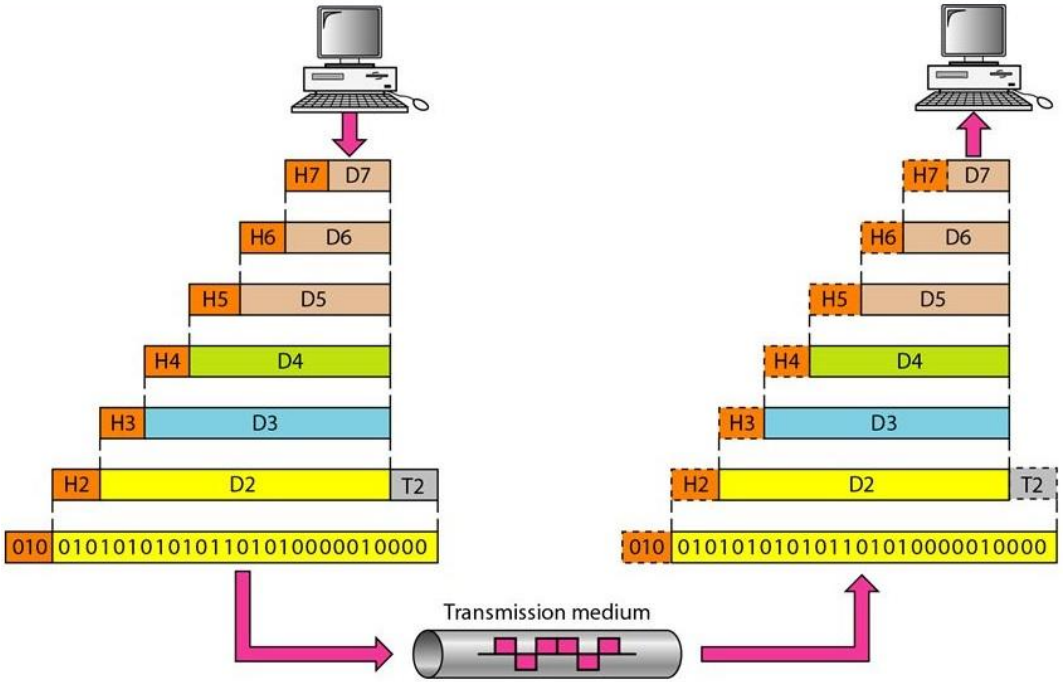
- Layers 1, 2, and 3 are the network support layers. They deal with the physical aspects of moving data from one device to another (such as specifications, physical connections, physical addressing, and transport timing and reliability).

- Layers 5,6, and 7 are the user support layers. They allow interoperability among unrelated software systems.

- Layer 4 links the two subgroups and ensures that what the lower layers have transmitted is in a form that the upper layers can use.

Headers and trailers

H2: physical address of the most recent node and next intended node;
T2: error control information;
H3: source and destination addresses;
H4: sequence number;
H5: connection control information (password and login verification);
H6: type and length of the transmission (type of code, etc.)



An exchange using the OSI model

What's a protocol?

human protocols:

- "what's the time?"
- "I have a question"
- introductions

… specific msgs sent

… specific actions taken when msgs received, or other events

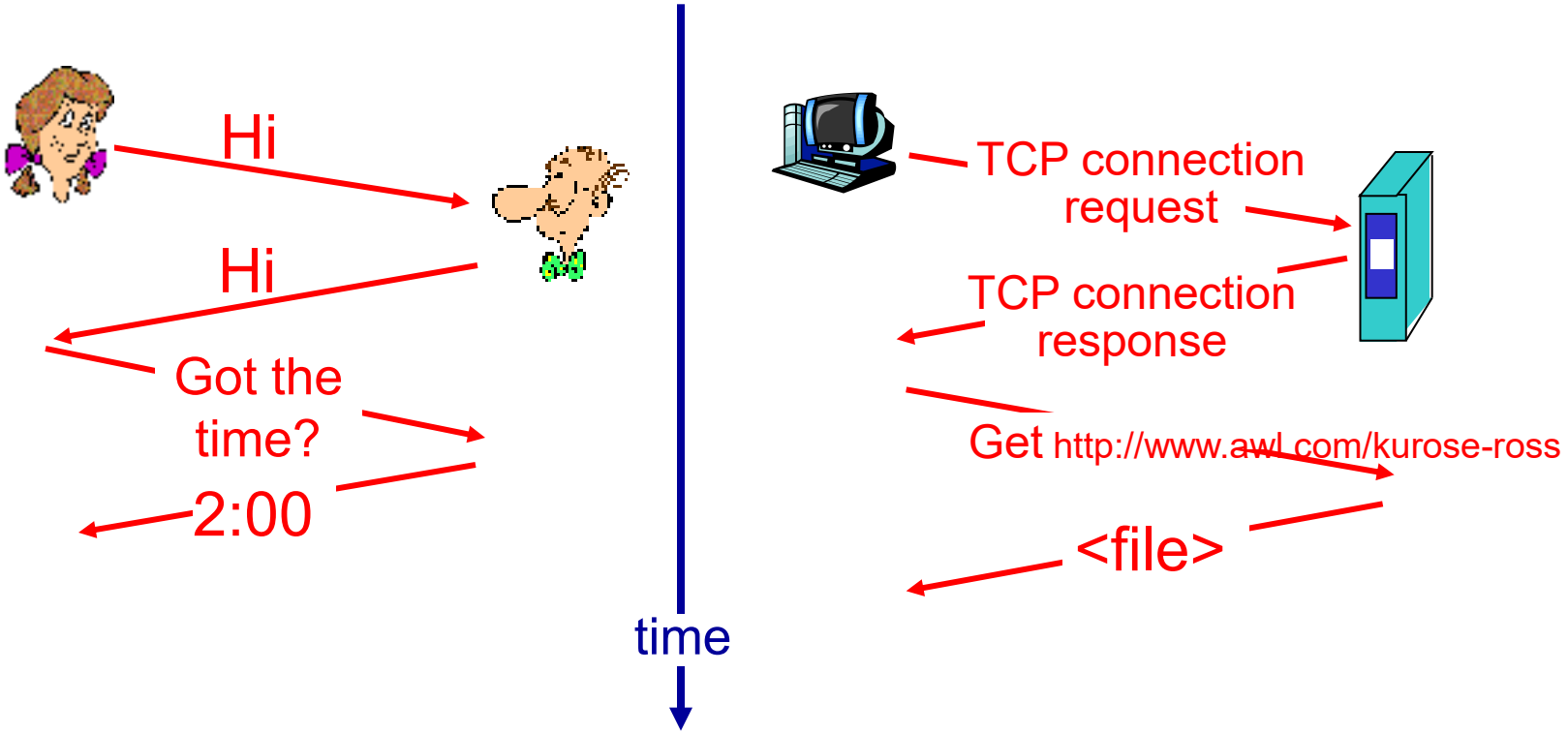
network protocols:

- machines rather than humans
- all communication activity in Internet governed by protocols

protocols define format, order of msgs sent and received among network entities, and actions taken on msg transmission, receipt

What's a protocol?

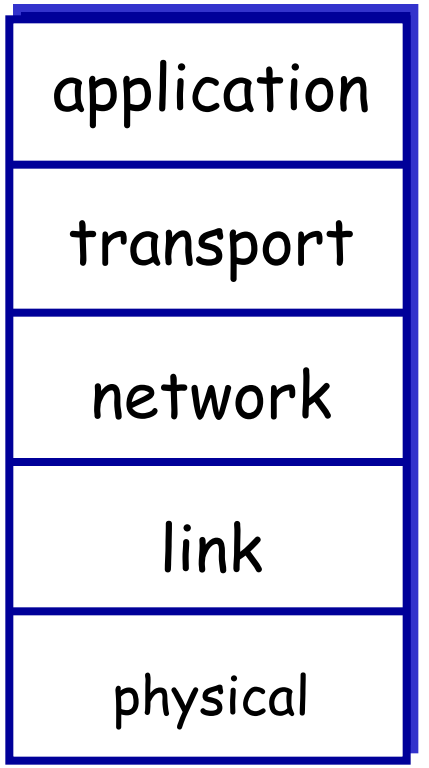
a human protocol and a computer network protocol:



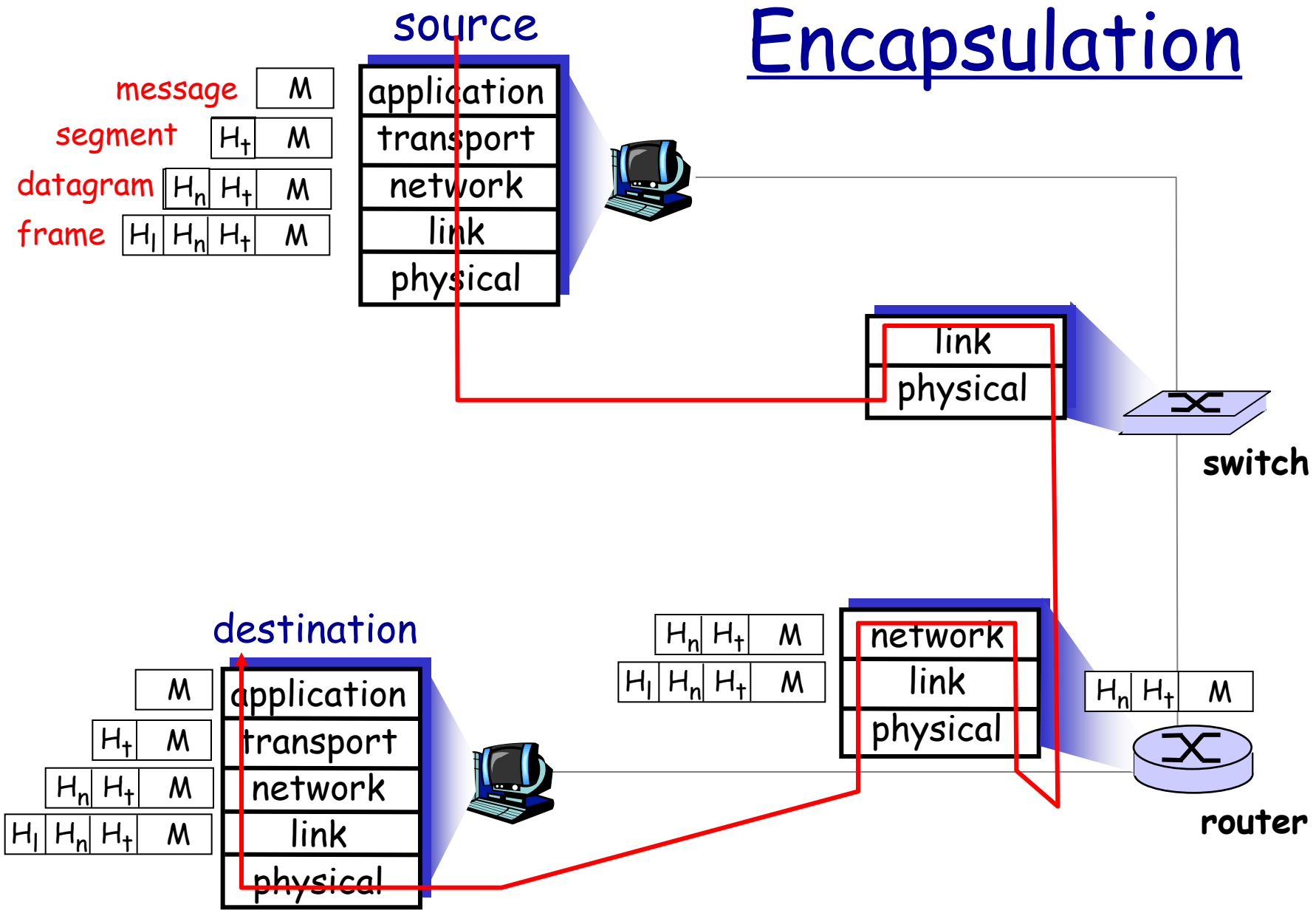
Q: Other human protocols?

TCP/IP protocol stack

- application: supporting network applications
  - FTP, SMTP, HTTP
- transport: process-process data transfer
  - TCP, UDP
- network: routing of datagrams from source to destination
  - IP, routing protocols
- link: data transfer between neighboring network elements (MAC/Link)
- physical: bits "on the wire": Ethernet, 802.11 (WiFi)



Encapsulation



Chapter 1: Introduction

1.1 What *is* the Internet?

1.2 Network edge

- end systems, access networks, links

1.3 Network core

- circuit switching, packet switching, network structure

1.4 The way networks work

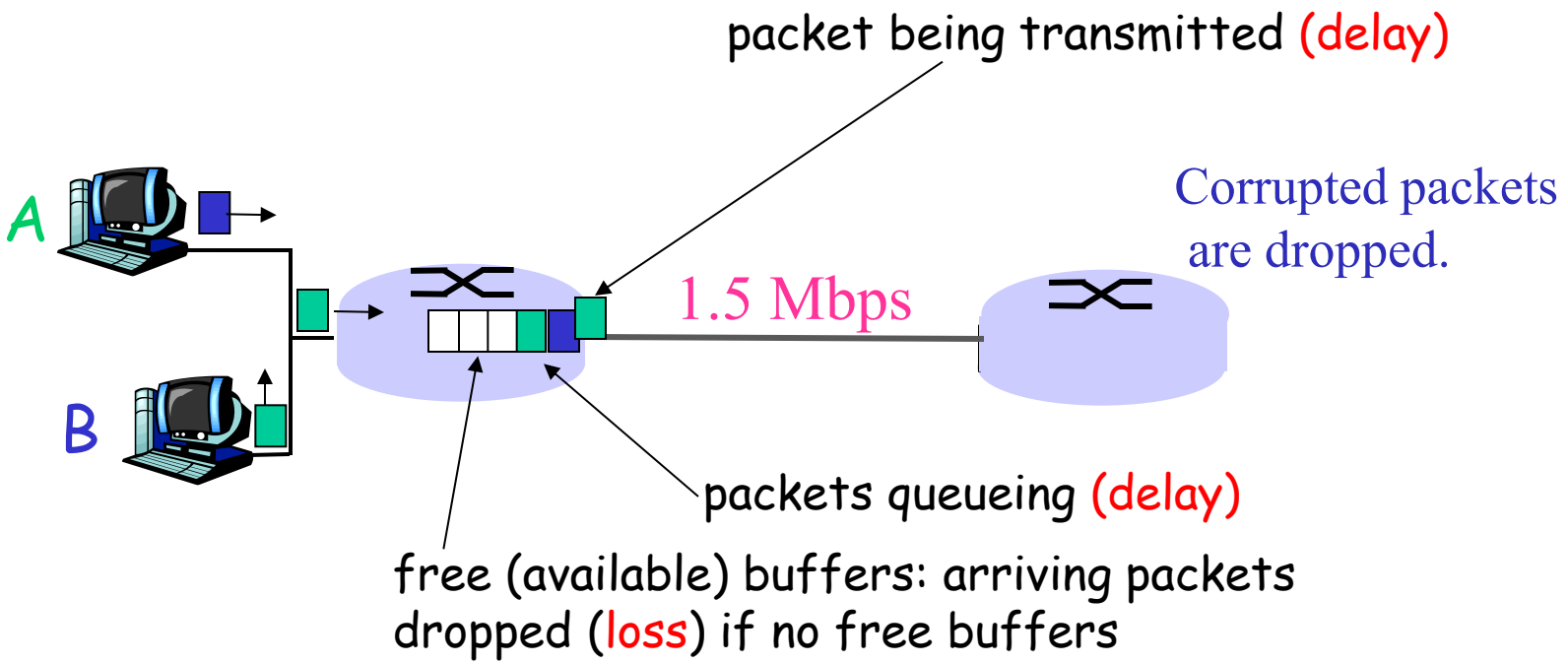
1.5 Performance metrics:

Delay, loss and throughput in packet-switched networks

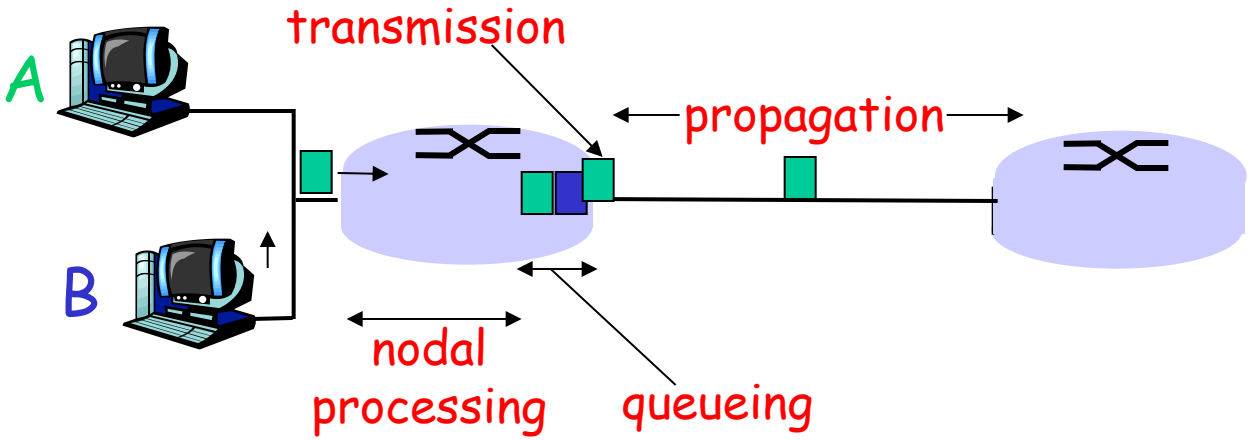
How do loss and delay occur?

packets queue in router buffers (i.e., memory)

- packet arrival rate to link exceeds output link capacity
- packets queue, wait for turn



Four sources of packet delay



$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

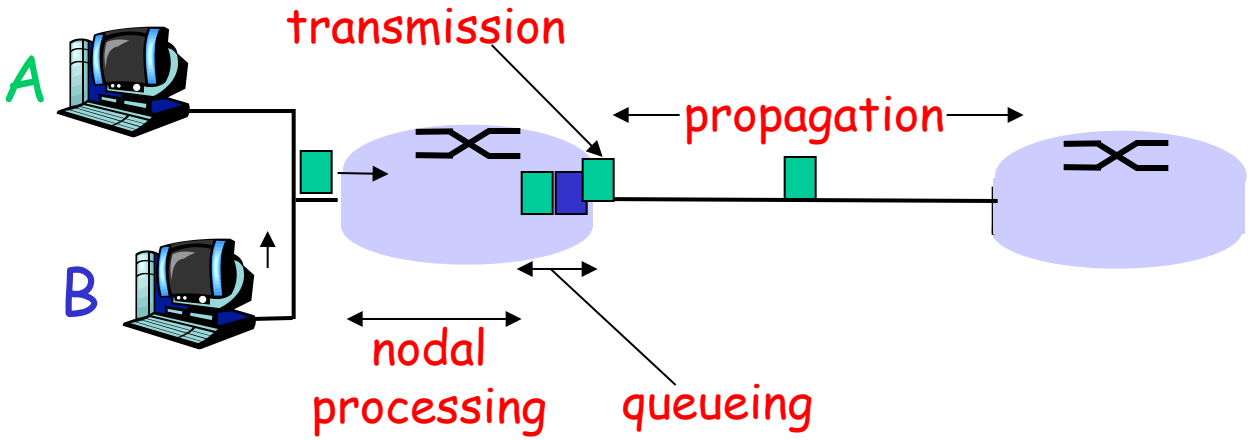
d_{proc}: nodal processing

- check bit errors
- determine output link
- typically < msec

d_{queue}: queueing delay

- time waiting at output link for transmission
- depends on congestion level of router

Four sources of packet delay



$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

d_{trans}: transmission delay:

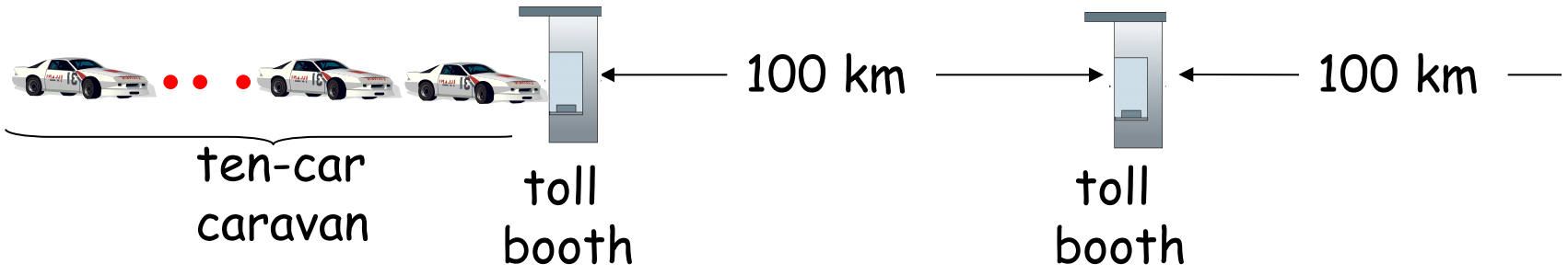
- L: packet length (bits)
- R: link bandwidth (bps)
- $d_{trans} = L/R$

d_{prop}: propagation delay:

- d: length of physical link
- s: propagation speed in medium (~2×10^8 m/sec)
- $d_{prop} = d/s$

d_{trans} and d_{prop} *very different*

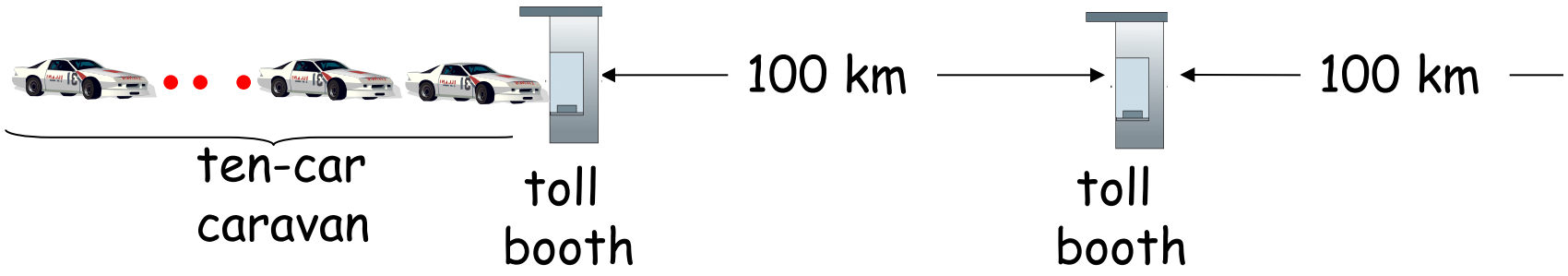
Caravan analogy



- cars "propagate" at 100 km/hr
- toll booth takes 12 sec to service car (transmission time)
- car~bit; caravan ~ packet
- Q: How long until caravan is lined up before 2nd toll booth?

- time to "push" entire caravan through toll booth onto highway = 12*10 = 120 sec
- time for last car to propagate from 1st to 2nd toll booth: 100km/(100km/hr)= 1 hr
- A: 62 minutes

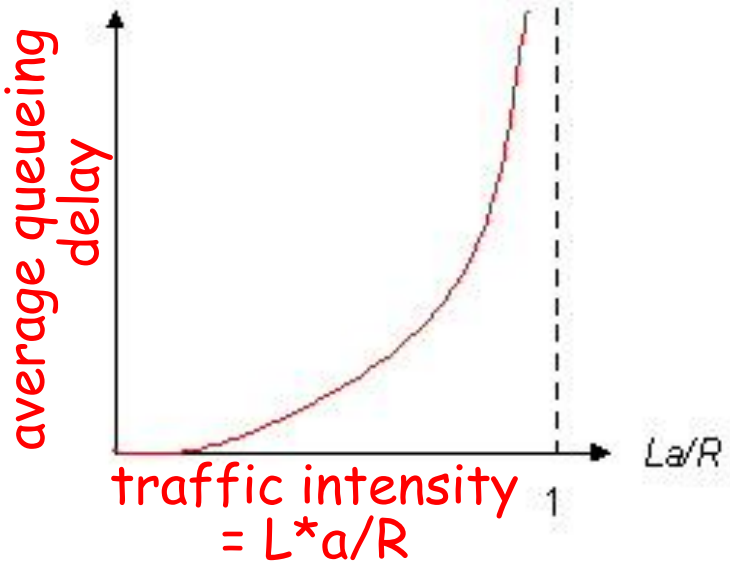
Caravan analogy



- cars now "propagate" at 1000 km/hr
- toll booth now takes 1 min to service a car
- Q: Will the first car arrive to 2nd booth before all cars serviced at 1st booth?
  - A: Yes! After 7 min, 1st car arrives at second booth; three cars still at 1st booth.
  - 1st bit of packet can arrive at 2nd router before packet is fully transmitted at 1st router!

Queueing delay

- R: link bandwidth (bps)
- L: packet length (bits)
- a: average packet arrival rate (#packets/s)



- La/R ~ 0: avg. queueing delay small
- La/R -> 1: avg. queueing delay large
- La/R > 1: more "work" arriving than can be serviced, average delay infinite!



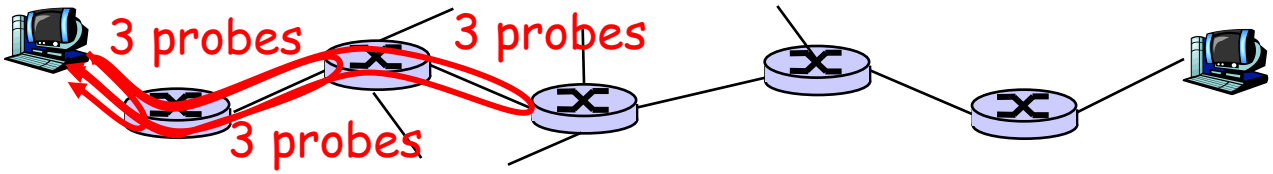
La/R ~ 0



La/R -> 1

"Real" Internet delays and routes

- What do "real" Internet delay & loss look like?
- Traceroute program: provides delay measurement from source to router along end-end Internet path towards destination. For all i:
  - sends three packets that will reach router *i* on path towards destination
  - router i will return packets to sender
  - sender times interval between transmission and reply.



C:xyz>tracert u-aizu.ac.jp

"Real" Internet delays and routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr

Three delay measurements from gaia.cs.umass.edu to cs-gw.cs.umass.edu

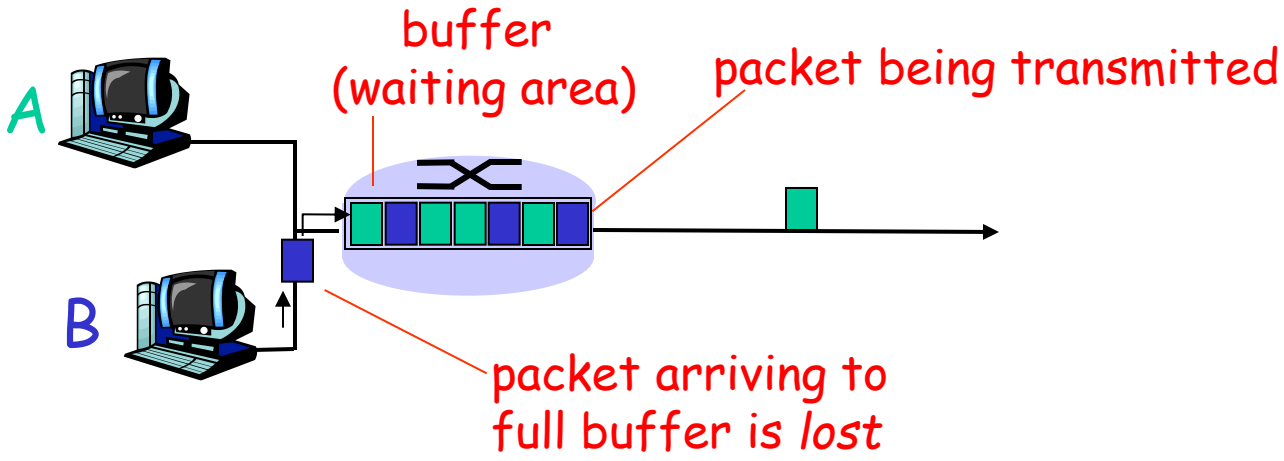
```
1  cs-gw (128.119.240.254)  1 ms  1 ms  2 ms
2  border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145)  1 ms  1 ms  2 ms
3  cht-vbns.gw.umass.edu (128.119.3.130)  6 ms 5 ms 5 ms
4  jn1-at1-0-0-19.wor.vbns.net (204.147.132.129)  16 ms 11 ms 13 ms
5  jn1-so7-0-0-0.wae.vbns.net (204.147.136.136)  21 ms 18 ms 18 ms
6  abilene-vbns.abilene.ucaid.edu (198.32.11.9)  22 ms  18 ms  22 ms
7  nycm-wash.abilene.ucaid.edu (198.32.8.46)  22 ms  22 ms  22 ms
8  62.40.103.253 (62.40.103.253)  104 ms 109 ms 106 ms
9  de2-1.de1.de.geant.net (62.40.96.129)  109 ms 102 ms 104 ms
10  de.fr1.fr.geant.net (62.40.96.50)  113 ms 121 ms 114 ms
11  renater-gw.fr1.fr.geant.net (62.40.103.54)  112 ms  114 ms  112 ms
12  nio-n2.cssi.renater.fr (193.51.206.13)  111 ms  114 ms  116 ms
13  nice.cssi.renater.fr (195.220.98.102)  123 ms  125 ms  124 ms
14  r3t2-nice.cssi.renater.fr (195.220.98.110)  126 ms  126 ms  124 ms
15  eurecom-valbonne.r3t2.ft.net (193.48.50.54)  135 ms  128 ms  133 ms
16  194.214.211.25 (194.214.211.25)  126 ms  128 ms  126 ms
17  * * *
18  * * *
19  fantasia.eurecom.fr (193.55.113.142)  132 ms  128 ms  136 ms
```

trans-oceanic link

* means no response (probe lost, router not replying)

Packet loss

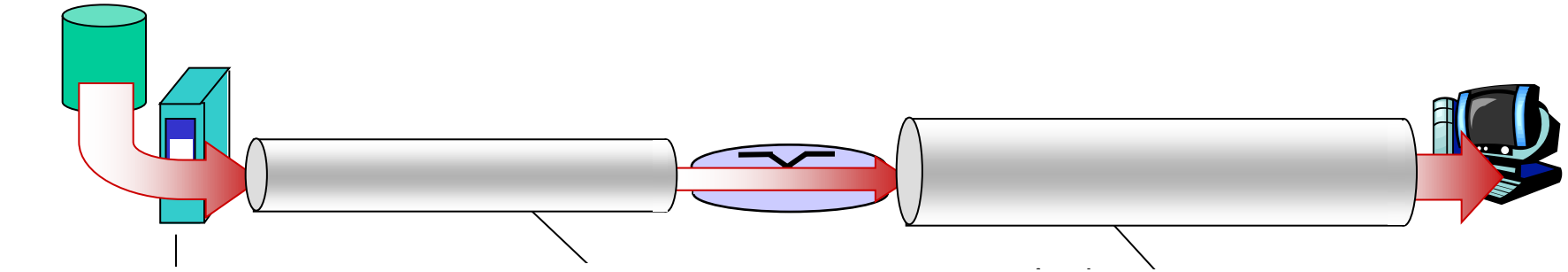
- queue (i.e., buffer) preceding link has finite capacity
- packet arriving to full queue dropped (lost)
- lost packet may be retransmitted by previous node, by source end system, or not at all



Throughput

(the output rate of an input/output system)

- throughput: rate (bits/time unit) at which bits are transmitted between sender/receiver
  - instantaneous: rate at given point in time (=max rate)
  - average: rate over longer period of time



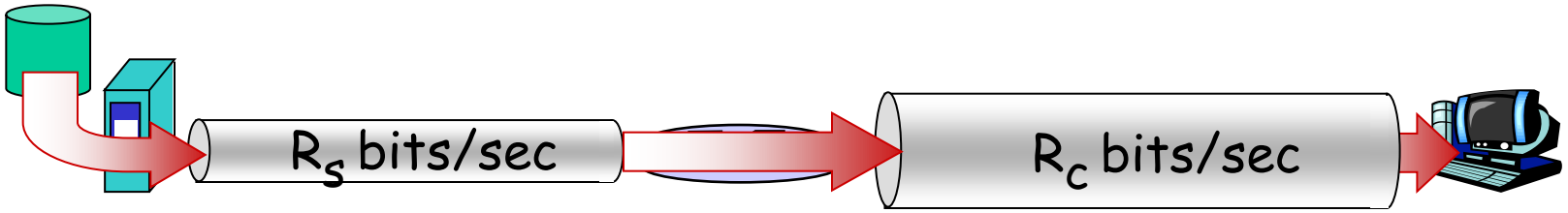
server sends bits (fluid) into pipe

pipe that can carry fluid at rate R_s bits/sec)

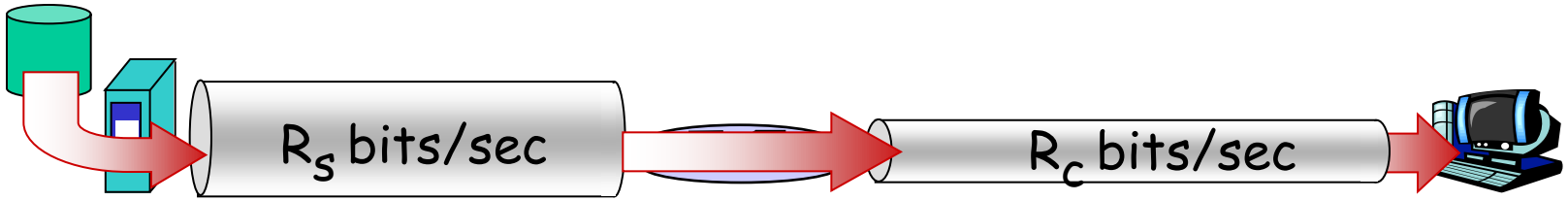
pipe that can carry fluid at rate R_c bits/sec)

Throughput

- $R_s < R_c$ What is average end-end throughput?



- $R_s > R_c$ What is average end-end throughput?

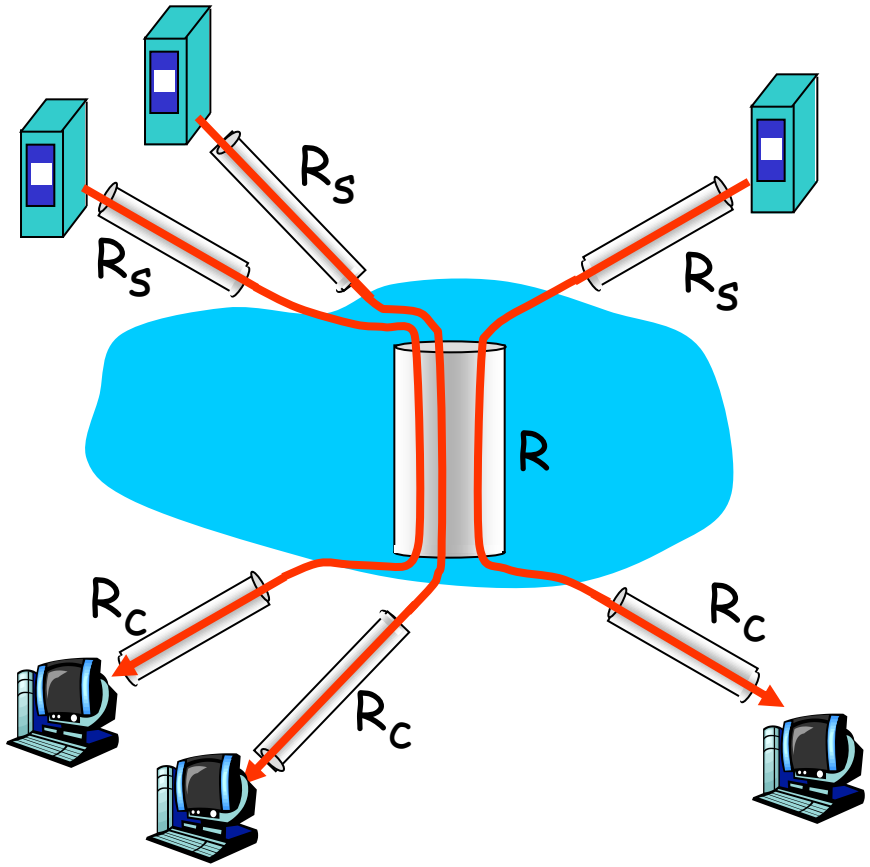


bottleneck link

link on end-end path that constrains end-end throughput

Throughput: Internet scenario

- per-connection end-end throughput: $\min(R_c, R_s, R/10)$
- in practice: R_c or R_s is often bottleneck



10 connections (fairly) share backbone bottleneck link R bits/sec